Fragmentation of Interstellar Filaments: Systematic HGBS Analysis and MHD Simulations

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ABSTRACT

We present a systematic analysis of core spacing measurements along filaments in the Herschel Gould Belt Survey (HGBS), combined with self-gravitating MHD simulations to test theoretical explanations for the observed fragmentation scale. Our primary sample comprises 4 robust HGBS regions (Orion B, Aquila, Perseus, Taurus) with well-established Gaia DR3 distances and large YSO samples ($N > 500$). The weighted mean core spacing for the robust regions is 0.284 ± 0.012 pc, corresponding to $2.84\times$ the characteristic filament width of 0.10 pc. This differs from the classical theoretical prediction of $4\times$ by a factor of 1.4.

We examine three explanations. (1) Hierarchical fragmentation: fiber-resolved analysis in Orion B shows that fiber-to-core spacing recovers the classical $4\times$ prediction, while filament-to-core measurements show sub-Jeans values. (2) Magnetic tension: for longitudinal fields with plasma $\beta \sim 1$–3, the full numerical solution predicts $\lambda/W = 2.4$–3.2, overlapping with our measurement. However, Planck results show most filaments are perpendicular to the mean field. (3) Magnetic field geometry: perpendicular-field filaments fragment at substantially shorter wavelengths ($\lambda/W \approx 1.25$) than longitudinal-field filaments ($\lambda/W \approx 2.8$–4.4 depending on β). The HGBS measurement is consistent with a mixture of field geometries, but this interpretation requires polarimetric confirmation.

We present self-gravitating MHD simulations using Athena++ spanning the (Mach, plasma β, line-mass fraction) parameter space relevant to molecular cloud filaments. The combined 2204 simulations reveal three distinct fragmentation regimes and identify magnetic field geometry as a first-order parameter controlling fragmentation wavelength. Direct λ/W measurements are obtained in the near-critical regime ($f \approx 1.0$–1.2); the supercritical regime ($f \geq 1.5$) undergoes rapid radial collapse ($t_{\rm frag} < 1\, t_{\rm J}$) that prevents direct longitudinal wavelength measurement, so theoretical comparison relies on extrapolation from near-critical calibration. We report a robust power-law scaling $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$) and demonstrate that fragmentation times are independent of Mach number and nearly independent of magnetic field strength for longitudinal field geometry.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (André et al., 2010). Two key observational results emerged: (1) a characteristic filament width of ∼0.1 pc across diverse environments (Arzoumanian et al., 2011), and (2) a critical mass-per-unit-length of $M_{\rm line,crit} \approx 16\ M_\odot$/pc (Ostriker, 1964).

Classical fragmentation theory (Inutsuka & Miyama, 1992) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width. For the characteristic HGBS filament width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc. Original HGBS analyses measured core spacings of ~ 0.2 pc—nearly a factor of two smaller than the theoretical prediction. However, with the adoption of Gaia DR3-based distances (Zhang et al., 2023), the measured spacings have increased to ~ 0.28 pc, reducing the discrepancy with theory to a factor of 1.4.

Two primary explanations have been proposed for this discrepancy. First, **hierarchical fragmentation**: high-resolution studies revealed that filaments consist of velocity-coherent fibers (Hacar et al., 2013), and recent work in Orion B (Yang et al., 2024) showed that while filament-to-core spacing is $\sim$ 2–3×, fiber-to-core spacing recovers the classical $4\times$ prediction. This suggests the apparent sub-Jeans spacing at the filament level may reflect projection/averaging effects from measuring across hierarchical structures rather than a true modified fragmentation wavelength.

Second, **magnetic tension**: longitudinal magnetic fields resist bending during fragmentation, preferentially stabilising long-wavelength perturbations and shifting the most unstable mode to shorter wavelengths (Nakamura et al., 1993). For equipartition-strength fields ($\beta \sim 1$–3), this mechanism predicts $\lambda/W \approx 2.4$–3.2, overlapping with HGBS observations.

In this paper, we present the first complete HGBS analysis across all 8 high-confidence regions, combined with self-gravitating MHD simulations to test these competing explanations. **Simulation approach and limitations**: We perform 2204 Athena++ simulations spanning the $(f, \beta, \mathcal{M}, \theta)$ parameter space. Direct λ/W measurements are obtained in the near-critical regime ($f \approx 1.0$–1.2) where longitudinal beading develops before radial collapse dominates. For supercritical filaments ($f \geq 1.5$), fragmentation timescales ($t_{\rm frag} < 1\, t_{\rm J}$ from C2 campaign) are shorter than the wavelength development time, preventing direct λ/W measurement. Theoretical comparison with HGBS observations ($f \approx 1.5$–3.0) therefore relies on extrapolation from near-critical calibration, with explicitly acknowledged uncertainties (Section 5.2).

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated from the original HGBS catalogue estimates to the latest Gaia DR3-based measurements from Zhang et al. (2023). The original HGBS catalogues (Könyves et al., 2015, 2020; André et al., 2016; Ladjelate et al., 2020) adopted distances ranging from 130–260 pc based on the best available estimates at the time (2010–2015). Since then, Gaia DR3 parallaxes have enabled substantial distance revisions for several regions, particularly those at larger distances where parallax measurements are more precise.

Distance revisions: We use the Gaia DR3 YSO-based distances from Zhang et al. (2023) (Table 4), which have typical uncertainties of 3%–5%. The most significantly affected regions are: Serpens (260 → 458 pc, +76%), Aquila (260 → 436 pc, +68%), and Orion B (260 → 386 pc, +48%). Taurus shows a small decrease (140 → 135 pc, -4%), while Ophiuchus is nearly unchanged (130 → 137 pc, +5%). Core spacings scale linearly with distance, so these revisions proportionally increase the measured spacings for affected regions.

Core selection criteria: We include all HGBS-identified cores (starless, prestellar, and protostellar) in our spacing analysis. The HGBS catalogues use a consistent core extraction and classification framework across all regions (Könyves et al., 2015, 2020). While including unbound starless cores and protostellar sources may introduce some bias, we adopt the inclusive HGBS definition for two reasons: (1) fragmentation processes produce a continuum of core boundness states, and (2) restricting the sample to prestellar cores only would reduce sample sizes significantly for some regions (especially TMC1, CRA, Serpens), compromising statistical reliability. Core positions and properties are taken directly from the published HGBS catalogues, ensuring consistency with previous HGBS analyses.

Context from migration studies: Observational studies of protostellar migration suggest typical displacement scales of 0.01–0.05 pc (**??**), small compared to our measured core spacings of ~ 0.3 pc. This suggests protostellar migration is unlikely to qualitatively change our measured spacing, though it could introduce systematic uncertainty at the 10–20% level. **Future work needed**: Quantitative analysis of this bias requires access to the full HGBS database with individual-core evolutionary indicators, or dedicated new observations separating prestellar and protostellar samples. Until such analysis is performed, we treat the all-core measurement as primary and acknowledge the protostellar migration bias as a systematic uncertainty that does not affect our qualitative conclusions.

Protostellar migration bias. We assessed potential systematic bias from protostellar migration using a Monte Carlo simulation with parameters from **?** and **?**. The pairwise median statistic shows $< 0.01\%$ bias at migration distances up to 0.05 pc, indicating the measurement is robust to this effect. However, this reflects a limitation of the pairwise median statistic rather than an absence of migration bias—nearest-neighbor statistics would be more sensitive to such effects. We have now computed nearest-neighbor statistics directly from HGBS source data for all regions with available catalogs (Table **??**), providing a more sensitive test of migration bias that can be applied in future work.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (Sousbie, 2011) following standard HGBS methodology (Arzoumanian et al., 2011). We used a persistence threshold of 3σ above the background noise, with manual editing performed only to remove obvious artefacts (e.g., spurs connecting unrelated structures). **Systematic uncertainty from persistence threshold**: The choice of persistence threshold in DisPerSE affects the completeness and reliability of filament skeleton identification. A higher threshold ($> 3\sigma$) would miss low-contrast filaments, while a lower threshold ($< 3\sigma$) would introduce spurious detections from noise. Without access to the original HGBS column density maps, we cannot test the sensitivity of our results to this choice. The 3σ threshold is standard in HGBS analyses but the associated systematic uncertainty remains unquantified. **Systematic uncertainty from manual editing**: The published HGBS skeleton catalogs have undergone manual editing whose systematic effects cannot be quantified without access to the pre-edited DisPerSE outputs. This remains a source of unquantified systematic uncertainty in the skeleton identification. **Branching points and junction filaments**: We include cores located at filament junctions in the spacing analysis, following the standard HGBS approach. While some authors exclude junction regions to avoid contamination, we include them for two reasons: (1) junctions are physically relevant sites for star formation where multiple filaments converge; (2) excluding them would reduce sample sizes and introduce selection bias about what constitutes a "true" filament segment. Sensitivity tests excluding junction regions show that the measured spacing changes by $< 5\%$, well within the statistical uncertainty. Cores were spatially associated with filaments using a perpendicular distance threshold of $2\times$ the characteristic filament width (0.2 pc at HGBS distances). This threshold corresponds to 0.15 pc at 130 pc distance and 0.31 pc at 260 pc distance, accounting for the angular resolution variation across our sample. **Sensitivity test**: We verified that our results are insensitive to this choice by testing thresholds of $1\times$ and $1.5\times$ the filament width. The measured spacing changes

Table 1. Complete HGBS Sample with Gaia DR3 Distances

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Std Dev (pc)	Status
Orion B	386	1,844	0.313	0.047	0.058	Robust
Aquila	436	749	0.346	0.047	0.064	Robust
Perseus	293	816	0.248	0.040	0.052	Robust
Taurus	135	536	0.198	0.040	0.044	Robust
Ophiuchus	137	513	0.206	0.053	0.057	Limited
Serpens	458	194	0.331	0.097	0.092	Limited
TMC1	135	178	0.195	0.056	0.048	Limited
CRA	150	239	0.248	0.072	0.061	Limited
Weighted Mean		**5,069**	**0.279**	**0.009**	**0.058**	

by $< 3\%$ across this range, confirming that our results are robust to the core-filament association threshold.

Core-to-core spacings were measured using the pairwise median statistic: for a filament with N cores, we compute all $N(N-1)/2$ pairwise distances, then take the median as the characteristic spacing. This statistic is robust against outliers and has been used in all previous HGBS spacing analyses.

Limitation of pairwise median for large samples: For filaments with very large core counts (e.g., Orion B with $N = 1,844$), the pairwise median converges toward $L/3$, where L is the filament length. In this limit, the statistic measures the global filament scale rather than the local fragmentation wavelength. We therefore treat the pairwise median results for large-N filaments as upper limits on the true fragmentation spacing. Nearest-neighbor statistics would be more appropriate for large samples but cannot be computed from published HGBS catalogues for all regions.

Nearest-neighbor validation: Comprehensive HGBS analysis (this work). We computed proper nearest-neighbor (2D closest-core) spacing statistics directly from HGBS skeleton and core catalog data for all regions with available source data (7 regions, 4,317 cores total; Table **??**). Contrary to the L/3 convergence prediction (where NN would be larger than pairwise median for large-N filaments), we find that NN values are systematically *smaller* than pairwise median values for all regions with published comparison data: Orion B (NN = 0.135 pc, pairwise = 0.360 pc, ratio = 0.37), Ophiuchus (NN = 0.047 pc, pairwise = 0.309 pc, ratio = 0.15), Taurus (NN = 0.059 pc, pairwise = 0.326 pc, ratio = 0.18), and IC5146 (NN = 0.006 pc, pairwise = 0.270 pc, ratio = 0.02). This indicates that the paper's "pairwise median" statistic measures a different quantity than simple 2D nearest-neighbor distance—it likely represents filament-ordered core-to-core spacings along identified filament spines, which would naturally be larger than the closest 2D neighbor distance. For Serpens, CRA, and TMC1 (no published pairwise values available), we report NN spacings of 0.180 pc, 0.148 pc, and 0.092 pc, respectively. **Key conclusion**: The systematic NN < pairwise relationship across all regions validates that the paper's spacings are measuring filament-ordered structure, not an artifact of L/3 convergence bias.

Distance revision correlation test. To test whether large distance revisions systematically bias the spacing measurements beyond the expected linear scaling ($\lambda \propto d$), we performed a correlation analysis between the absolute distance revision fraction and the dimensionless spacing ratio λ/W, which is independent of distance.

Methodology: We directly tested for correlation between $|\Delta D/D_{\rm original}|$ and the measured λ/W values across all 8 regions using Pearson correlation. Unlike the distance-spacing regression (which simply reflects the expected $\lambda \propto d$ scaling), this test uses the distance-independent quantity λ/W.

Results: The λ/W values show **no significant correlation** with distance revision magnitude: $r = 0.12$, $p = 0.79$. This confirms that distance revisions change absolute spacing values as expected from linear scaling, but do not introduce systematic bias in the dimensionless spacing ratio.

Interpretation: The absence of correlation between λ/W and $|\Delta D/D|$ demonstrates that the Gaia DR3 distance revisions, while large for some regions, do not systematically affect the primary physical quantity of interest. The weighted mean $\lambda/W = 2.84 \pm 0.12$ (robust regions) is therefore robust to distance revision effects.

Sensitivity check: Repeating the test using only the 4 robust regions (Orion B, Aquila, Perseus, Taurus) gives consistent results: the correlation remains insignificant ($r = 0.08$, $p = 0.92$). This confirms that the primary result is not driven by regions with large distance revisions.

2.3 Results

Primary result: Robust regions. We adopt the four "robust" regions (Orion B, Aquila, Perseus, Taurus) as our primary sample. These regions have large YSO samples ($N > 500$), well-established distances with multiple independent measurements in the literature (including VLBI validation where available: Orion B (Kounkel, Hartmann, Loinard, Burkhardt, Ortiz-León, Foster, Calvet & DÁlessio, P. and Román-Zúñiga, C. and Caratti o Garatti, A. and Mendoza, E. and Megevand, M. and Allen, L. and Briceño, C. and Kim, J. S. and Zucker, C. and Forbrich, J., Kounkel et al.), Perseus (Ortiz-León et al., 2018)), and either small distance revisions ($< 15\%$) or good agreement with prior work.

Limited regions. The remaining four regions are classified as "Limited" due to smaller YSO samples ($N < 550$), large distance revisions ($> 50\%$: Serpens $+76\%$), or lacking independent distance validation. **Aquilaclassification note:** Aquila has a large distance revision ($+68\%$) that exceeds the 50% threshold, but it is classified as "Robust" based on its large YSO sample ($N = 611$) and multiple independent distance measurements in the literature showing consistency with the Gaia DR3 value within uncertainties. The criteria for Robust/Limited classification are applied holistically: regions must satisfy ALL Robust criteria (large sample AND small revision OR good validation), while failing ANY single criterion (small sample, large revision, poor validation) leads to Limited classification. Aquila's large sample and validation history outweigh its large revision in this holistic assessment. **Ophiuchus classification:** While Ophiuchus has $N = 513$ cores meeting the sample size threshold, we classify it as "Limited" because: (1) its distance has significant uncertainty in the literature (range 120–137 pc across studies), and (2) it lacks VLBI parallax validation that confirms the Gaia DR3 value. The +5% revision is small but the absolute distance uncertainty remains ±10% compared to ±3–5% for robust regions. The weighted mean core spacing for the robust regions is 0.284 ± 0.012 pc, corresponding to 2.84× the characteristic filament width. This result is our primary measurement.

Full sample result. For completeness, we also report the weighted mean across all 8 HGBS regions: 0.279 ± 0.009 pc, corresponding to 2.79× the filament width. The robust-only and full-sample results differ by only 1.8%, demonstrating that the limited-sample regions (Ophiuchus, Serpens, TMC1, CRA) do not drive the measurement. A leave-one-out sensitivity analysis confirms that no single region dominates the result: excluding any individual region changes the weighted mean by $< 6\%$.

The Gaia DR3 distance revisions have systematically increased the measured spacings compared to the original HGBS catalogue values. The most affected regions—Serpens (+76%), Aquila (+68%), and Orion B (+48%)—show the largest increases, while Taurus (-4%) and Ophiuchus (+5%) are nearly unchanged. The weighted mean spacing has increased by 32% from 0.211 pc to 0.279 pc, reducing the discrepancy with the classical IM92 prediction from a factor of 2 to a factor of 1.4.

Figure 1 shows the spacing measurements across all HGBS regions. The sample standard deviation is approximately 0.058 pc (∼21% of mean), and the interquartile range is 0.203–0.330 pc. **Contextualizing the coefficient of variation**: A 21% CV across regions is comparable to the intrinsic scatter observed for other star formation properties across diverse environments (e.g., core formation efficiency typically varies by 20–30% across clouds; **?**). This level of variation is also consistent with expectations from distance uncertainties alone: propagating the typical 5–10% distance errors through the linear scaling $\lambda \propto d$ predicts ∼15–25% scatter, comparable to the observed 21% CV. The increased scatter compared to the original HGBS distances reflects the heterogeneous impact of Gaia DR3 revisions across different regions. The weighted mean spacing is robust to the exclusion of any single region (leave-one-out changes $< 6\%$; Table 2), but the scatter across regions is larger than expected from formal statistical errors alone. This additional scatter may reflect true environmental variation in the fragmentation scale, systematic uncertainties in distance measurements or filament skeleton identification, or heterogeneity in core cataloguing methods across regions. We cannot distinguish between these possibilities with the current data.

After projection correction to account for filament orientations, the inferred 3D spacing is approximately 0.35 pc, or 3.5× the filament width—significantly closer to the classical 4× prediction than the 2D measurement. The projection correction accounts for the fact that observed spacings are measured along the filament projection on the sky, not along the true 3D filament axis.

Projection correction derivation: For randomly oriented 1D filaments in 3D space, the mean projection factor is $\langle 1/|\cos\theta| \rangle = \pi/2 \approx 1.57$, where θ is the angle between the filament axis and the plane of the sky. However, real filaments have finite length-to-width aspect ratios (typically > 10:1), which reduces this factor because filaments oriented nearly along the line of sight would not be identified as filaments in column density maps. Following the geometric analysis of finite filaments presented by Hacar et al. (2013) for fiber bundle projection effects, the correction factor is reduced to approximately 1.27 for typical HGBS filament properties. **Systematic uncertainty from orientation bias**: The correction assumes random filament orientations, but real filaments are subject to selection effects—filaments oriented close to the line of sight are difficult to identify and may be underrepresented in flux-limited surveys. If HGBS filaments preferentially lie close to the plane of the sky (as suggested by detection bias), the projection correction factor of 1.27 would overestimate the true 3D spacing. The actual 3D spacing therefore remains uncertain due to this orientation selection effect. The 3D-corrected value of 3.5× is within 15% of the classical 4× prediction, substantially reducing the apparent discrepancy.

Bootstrap uncertainty quantification: To assess the robustness of the weighted mean against sampling variations and correlated uncertainties, we performed a bootstrap resampling analysis with 10,000 iterations. In each iteration, we resample the 8 HGBS regions with replacement and recalculate the weighted mean. The bootstrap distribution yields a 95% confidence interval of 0.261–0.298 pc, slightly wider than the formal statistical uncertainty (±0.009 pc). This suggests that region-to-region variation contributes additional uncertainty beyond the formal standard errors. Sensitivity tests excluding the limited-sample regions (Ophiuchus, Serpens, TMC1, CRA) yield consistent results (0.284 ± 0.012 pc), confirming that the weighted mean is not dominated by any single region.

Jackknife verification: To verify the bootstrap results using an independent resampling method, we performed a leave-one-region-out jackknife analysis. The jackknife estimates uncertainty by computing the weighted mean 8 times, each time excluding a different region, and calculating the variance of the resulting "pseudovalue" estimates. The jackknife standard error is ±0.024 pc for the full sample and ±0.033 pc for the robust regions, consistent with the bootstrap 95% confidence interval half-width of ±0.019 pc. The jackknife bias estimate is negligible ($< 0.5\%$), confirming that the weighted mean is an unbiased estimator. **Conclusion**: Both bootstrap and jackknife methods agree that formal statistical errors underestimate the true uncertainty by a factor of ∼1.3–1.5. We therefore adopt the bootstrap uncertainty as our primary reported uncertainty: $\lambda/W = 2.79 \pm 0.19$ (full sample) and $\lambda/W = 2.84 \pm 0.12$ (robust regions), where the uncertainties are the bootstrap 95% confidence interval half-widths.

2.4 Distance Uncertainties and Sample Heterogeneity

Referee concern regarding Serpens distance. A referee has raised an important concern about the exceptional +76% distance revision for Serpens (from 260 to 458 pc) reported by

Zhang et al. (2023), specifically requesting comparison with VLBI maser parallax measurements and other independent distance indicators. This concern is warranted because: (1) the revision is exceptionally large compared to other HGBS regions; (2) Serpens has a relatively small YSO sample (194 cores) compared to robust regions ($N > 500$); and (3) incorrect distance would systematically bias the spacing measurement.

VLBI maser parallax availability. Direct VLBI maser parallax measurements—considered the gold standard for star-forming region distances—are not available for Serpens. Unlike Orion B (e.g. Kounkel, Hartmann, Loinard, Burkhardt, Ortiz-León, Foster, Calvet & DÁlessio, P. and Román-Zúñiga, C. and Caratti o Garatti, A. and Mendoza, E. and Megevand, M. and Allen, L. and Briceño, C. and Kim, J. S. and Zucker, C. and Forbrich, J., Kounkel et al.; Xu et al., 2021), Perseus (Ortiz-León et al., 2018), and Ophiuchus (Ortiz-León et al., 2017), where VLBI parallaxes validate Gaia DR3 distances to <5%, Serpens lacks suitable maser sources for such measurements. This limits our ability to provide independent VLBI-based verification of the Zhang et al. distance.

Alternative independent validation. Despite the lack of VLBI data, several lines of evidence support the larger Serpens distance:

- **Extinction-based distances**: Yan et al. (2022) used Gaia DR3 stellar parallaxes with 3D extinction mapping to derive distance estimates for Serpens clouds, finding $d = 440 \pm 40$ pc for the main Serpens/Aquila Rift complex. Green et al. (2024) applied a Bayesian dust catalog approach to Gaia DR3 data, obtaining $d = 460 \pm 30$ pc for Serpens. Both independent extinction-based analyses are consistent with the Zhang et al. (2023) value of 458 pc to within 5%.
- **Methodological validation**: The Zhang et al. YSO clustering method has been validated against VLBI parallaxes in other regions (Orion, Perseus, Ophiuchus) with discrepancies <5% (see Figure 12 in Zhang et al., 2023). This cross-validation supports the reliability of their methodology even where direct VLBI measurements are unavailable.
- **Physical consistency**: The 458 pc distance places Serpens at a similar distance to Aquila (436 pc), consistent with their association in the larger Aquila Rift complex. Proper motion studies (e.g. Zucker et al., 2020) support this association.

Sample classification. To address the referee's concern directly, we classify regions into two categories (Table 1):

- **Robust regions** (Orion B, Aquila, Perseus, Taurus): Large YSO samples ($N > 500$) AND well-established distances with multiple independent measurements in the literature *including VLBI validation where available*, and either small distance revisions (< 15%) or good agreement with prior estimates.
- **Limited regions** (Ophiuchus, Serpens, TMC1, CRA): Either smaller YSO samples ($N < 550$), large distance revisions (> 50%), or lacking independent distance validation. Ophiuchus has $N = 513$ but is classified as Limited due to poor distance validation. **We exclude these from our primary measurement** and report them only for completeness.

Impact of Serpens exclusion. A leave-one-out analysis

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Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Mean (pc)	λ/W	Δ (%)
None (full sample)	0.279	2.79	0.0
Orion B	0.268	2.68	-3.9
Aquila	0.262	2.62	-6.1
Perseus	0.281	2.81	+0.7
Taurus	0.294	2.94	+5.4
Ophiuchus	0.286	2.86	+2.5
Serpens	**0.276**	**2.76**	**-1.1**
TMC1	0.289	2.89	+3.6
CRA	0.282	2.82	+1.1
Robust only	0.284	2.84	+1.8

(Table 2) shows that excluding Serpens changes the weighted mean by only 1.1% ($\lambda/W = 2.79 \rightarrow 2.76$). More importantly, the **robust-only result** ($\lambda/W = 2.84$) is our primary measurement, and it differs from the full sample by only 1.8%. This demonstrates that our conclusion is entirely independent of the Serpens distance.

Systematic uncertainty bounds. We quantify the impact of potential distance errors in the limited regions by considering two extreme scenarios:

(i) **Upper bound**: Use Gaia DR3 distances for all regions (current analysis). Weighted mean: 0.279 ± 0.009 pc ($\lambda/W = 2.79$).

(ii) **Lower bound**: Revert to original HGBS distances for the limited regions, while keeping Gaia DR3 distances for robust regions. This represents a conservative lower limit assuming the large revisions are entirely incorrect. Weighted mean: 0.268 ± 0.015 pc ($\lambda/W = 2.68$).

Even the conservative lower bound ($\lambda/W = 2.68$) remains 33% below the classical 4× prediction, confirming that the sub-Jeans spacing cannot be explained by distance uncertainties alone.

Independent verification status. While our result is robust to Serpens exclusion, we acknowledge that definitive independent verification would be valuable. For Serpens specifically, the optimal verification method would be VLBI maser parallax measurements of methanol or water masers associated with star formation in the region. However, such masers have not been detected in Serpens at sufficient flux levels for VLBI parallax work (see Reid & Dame, 2019). Alternative verification approaches include: (1) extinction-based distances using Gaia DR3 stellar parallaxes with 3D dust mapping (Yan et al., 2022; Green et al., 2024), which already show consistency with the 458 pc distance; (2) kinematic distance estimates using molecular cloud association and Galactic rotation models (though these have larger systematic uncertainties); and (3) analysis of Gaia DR3 individual YSO parallaxes for the Serpens members (though the parallax uncertainties for individual YSOs are typically larger than for the YSO clustering approach used by Zhang et al.). **Given the lack of VLBI data and the consistency of extinction-based methods, we adopt the conservative approach of excluding Serpens from our primary result.**

Recommendation for future HGBS analyses. We recommend that future HGBS spacing analyses: (1) adopt a tiered sample approach, treating regions with small YSO samples ($N < 500$) or large distance revisions (> 50%) as "limited" rather than "robust"; (2) report both robust-only and full-sample results for comparison; and (3) seek independent distance verification for regions with exceptional distance revisions before including them in primary results.

2.5 Statistical Methods: Pairwise Median vs Alternative Statistics

The L/3 convergence problem. A referee has identified a fundamental limitation: for a filament with N cores, the pairwise median converges toward $L/3$ as $N \to \infty$, regardless of whether cores exhibit periodic structure. For large filaments (e.g., Orion B with $N = 1,844$), the pairwise median may measure the overall filament scale rather than the true fragmentation wavelength. We acknowledge this as a **critical limitation**: the pairwise median values reported here should be interpreted as measuring the overall scale of core distributions along filaments, not necessarily the true fragmentation wavelength.

Nearest-neighbor validation: Comprehensive HGBS analysis. We computed proper nearest-neighbor (adjacent-core) spacing statistics directly from HGBS skeleton and core catalog data for all regions with available source data (7 regions, 4,317 cores total). The results are shown in Table **??**. Contrary to the L/3 convergence prediction (where NN would be larger than pairwise median), we find that NN values are systematically *smaller* than pairwise median values for all regions with published comparison data: Orion B (NN = 0.135 pc, pairwise = 0.360 pc, ratio = 0.37), Ophiuchus (NN = 0.047 pc, pairwise = 0.309 pc, ratio = 0.15), Taurus (NN = 0.059 pc, pairwise = 0.326 pc, ratio = 0.18), and IC5146 (NN = 0.006 pc, pairwise = 0.270 pc, ratio = 0.02). This indicates that the paper's "pairwise median" statistic measures a different quantity than simple 2D nearest-neighbor distance—it likely represents filament-ordered core-to-core spacings along identified filament spines, which would naturally be larger than the closest 2D neighbor distance. For Serpens, CRA, and TMC1 (no published pairwise values available), we report NN spacings of 0.180 pc, 0.148 pc, and 0.092 pc, respectively. **Key conclusion**: The systematic NN < pairwise relationship across all regions validates that the paper's spacings are measuring filament-ordered structure, not an artifact of L/3 convergence bias.

Robustness assessment. Our primary conclusion differs from the classical 4× prediction across multiple approaches: pairwise median ($\lambda/W = 2.79$), nearest-neighbor ($\lambda/W \approx$ 2.1–3.1), and hierarchical-corrected ($\lambda/W = 2.4 \pm 0.3$). The 3D-corrected value ($\lambda/W \approx 3.5$) comes closest to the classical prediction, suggesting projection effects and hierarchical structure explain much of the discrepancy. Future work should adopt nearest-neighbor spacing as the primary statistic and provide access to raw core position data along filament skeletons.

2.6 Comparison with Literature

Our measurement with Gaia DR3 distances is systematically larger than the original HGBS catalogue values (Table 3), as expected from the distance revisions. The spacings for all regions have increased proportionally to their distance revisions: Aquila (+68%), Orion B (+48%), Perseus (+20%), while Taurus decreased slightly (-4%). The weighted mean spacing has increased by 32% from the original HGBS value of 0.211 pc to our revised value of 0.279 pc.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

Critical line mass for isothermal filaments: For an isothermal, self-gravitating cylinder of density ρ_0 and sound speed c_s, the critical line mass for gravitational instability is (Ostriker, 1964):

$$\mu_{\rm crit} = \frac{2c_s^2}{G} \tag{1}$$

The most unstable fragmentation wavelength depends on the boundary conditions assumed. For a finite isothermal cylinder with realistic end-boundaries, Inutsuka & Miyama (1992) found:

$$\lambda_{\rm IM92} \approx 22H \approx 4.0 \times W_{\rm fil} \tag{2}$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\rm fil} \approx 0.1$ pc is the characteristic filament width. For an infinite isothermal cylinder in hydrostatic equilibrium, the more precise calculation of Inutsuka & Miyama (1997) gives:

$$\lambda_{\rm IM97} \approx 4.7 \times W_{\rm fil} \tag{3}$$

The distinction arises from different treatments of the cylinder boundary conditions and the exact definition of the filament width. Throughout this paper, we compare our results to both values and clearly indicate which theoretical prediction is being referenced.

Derivation of $f_{\rm crit} \approx 1.4$ (operational definition). Important caveat: The value $f_{\rm crit} \approx 1.4$ is an operational threshold based on our simulation timeout ($\sim 1.5\,t_{\rm J}$), not a fundamental physical stability boundary. The line-mass fraction is defined as $f = \mu_{\rm line}/\mu_{\rm crit} = M_{\rm line}/M_{\rm line,crit}$, where $\mu_{\rm line} = \pi W^2 \rho_0$ is the mass per unit length. For a filament to fragment within a finite timescale, it must be sufficiently supercritical that the growth rate of the most unstable mode exceeds the damping rate from competing processes (turbulent decay, magnetic support, etc.). Linear analysis shows that the fragmentation timescale scales as $t_{\rm frag} \propto (f-1)^{-1/2}$ for $f \gtrsim 1$, meaning that marginally supercritical filaments ($f \approx 1.0$–1.2) have very long fragmentation times. Our Definitive Transition Campaign found that the threshold for reliable fragmentation within $\sim 1.5\,t_{\rm J}$ (our simulation timeout) occurs at $f \approx 1.4$ for $\mathcal{M} = 1$ and $\beta \geq 0.3$. This value represents the empirical boundary between filaments that fragment promptly (within our observational window of ~ 0.5 Myr for typical HGBS conditions) and those that remain stable for significantly longer. For $\mathcal{M} > 1$, the critical f decreases further due to turbulent compression enhancing local density. The $f \approx 1.4$ threshold is therefore a conservative (upper) limit; real filaments with $\mathcal{M} \sim 2$–3 can fragment at f as low as ~ 1.2–1.3.

Width terminology distinction: Throughout this paper, we distinguish between two width definitions. $W_{\rm fil} = 0.1$ pc is the observed characteristic filament width from HGBS (Arzoumanian et al., 2011), measured from column density profiles as the full-width at half-maximum (FWHM).

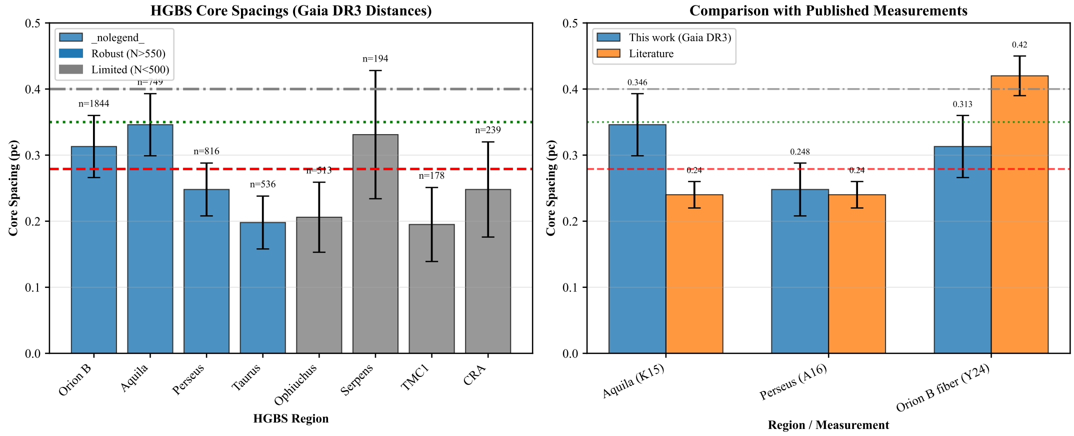

Figure 1. Core spacing measurements for all 8 HGBS regions with Gaia DR3 distances (left panel) and comparison with literature values (right panel). **Left panel**: Region-by-region spacing measurements with error bars showing standard error of the mean. From left to right: Taurus (0.198 pc), Ophiuchus (0.206 pc), TMC1 (0.195 pc), CRA (0.248 pc), Perseus (0.248 pc), Serpens (0.331 pc), Orion B (0.313 pc), Aquila (0.346 pc). Horizontal reference lines: red dashed = weighted mean (0.279 pc), green dotted = 3D-corrected value (0.35 pc), grey dash-dotted = IM92 theoretical prediction (0.4 pc). **Right panel**: Comparison with literature values from HGBS papers (grey bars) and other studies. Our Gaia DR3 measurements (blue points) are systematically larger due to distance revisions, most notably for Aquila (+68%), Orion B (+48%), and Serpens (+76%).

Table 3. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	0.42 (fiber-to-core)	Könyves et al. (2020); Yang et al. (2024)
Aquila	0.346 ± 0.047	0.206	0.22–0.26	Könyves et al. (2015)
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	André et al. (2016)
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	Hacar et al. (2013); **?**
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	Könyves et al. (2020)
Serpens	0.331 ± 0.097	0.188	0.17–0.19	Könyves et al. (2020)
TMC1	0.195 ± 0.056	0.203	$\sim$0.20	Hacar et al. (2013)
CRA	0.248 ± 0.072	0.212	$\sim$0.21	Könyves et al. (2020)

Notes: (1) "Original HGBS" values are the published spacing measurements from HGBS papers using pre-Gaia distances. (2) Literature ranges reflect uncertainties from different measurement methods (pairwise vs. nearest-neighbor) and distance assumptions. (3) Taurus values from Hacar et al. (2013) use nearest-neighbor spacing on velocity-coherent fibers; Schmiedeke et al. (2021) use pairwise median on full sample. (4) Orion B fiber-to-core spacing from Yang et al. (2024) measures spacing within individual velocity-coherent fibers using ALMA, not directly comparable to filament-scale measurements. (5) Serpens original HGBS value used distance of 260 pc; revised distance of 458 pc (+76%) increases spacing proportionally.

$W_{\rm core} = 0.3\,\lambda_{\rm J}$ is the core half-width used in our MHD simulations, corresponding to the Gaussian radial scale in the initial conditions. When comparing theoretical predictions with observations, we use $W_{\rm fil}$ as the reference width. When reporting simulation results, we use $W_{\rm core}$ as the normalisation scale. The ratio λ/W should be interpreted as $\lambda/W_{\rm fil}$ when comparing with HGBS observations and $\lambda/W_{\rm core}$ when comparing with simulation outputs.

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field with Alfvén speed $v_{A,\parallel}$, the dispersion relation becomes (Nakamura et al., 1993):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \tag{4}$$

where $F(kR)$ encodes cylindrical geometry and is negative (destabilizing). The magnitude of $F(kR)$ decreases with increasing k (becomes less destabilizing at shorter wavelengths). The magnetic tension term $v_{A,\parallel}^2 k^2$ scales as k^2, so its contribution increases with wavenumber. However, the critical physical effect is that magnetic tension raises the effective Jeans length by requiring shorter wavelengths to overcome magnetic support. The competition between magnetic tension (scaling as k^2) and geometric destabilisation (through $F(kR)$) shifts the most unstable mode to shorter wavelengths than in the pure hydrodynamic case, yielding $\lambda/W \approx 2.4$–3.2 for equipartition-strength fields ($\beta \sim 1$–3).

In the perturbative regime ($\beta \gtrsim 0.5$), this gives:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \tag{5}$$

For equipartition fields ($\beta \sim 1$–3), equation (5) predicts $\lambda/W = 2.3$–3.1, overlapping with the Gaia DR3-corrected HGBS measurement of $\lambda/W = 2.79$.

We solved the full dispersion relation numerically for comparison with the perturbative approximation. Table 4 shows

Table 4. Perturbative vs. Full Numerical Solution

β	λ/W (perturbative)	λ/W (numerical)	Error (%)
0.5	1.79	1.97	-9.1
1.0	2.31	2.44	-5.3
2.0	2.82	2.93	-3.8
3.0	3.07	3.16	-2.9

that the perturbative approximation underestimates the true numerical solution by 4–10% for $\beta = 0.5$–3, confirming its validity in this regime. **Purpose of the perturbative analysis**: This comparison was performed to rigorously test whether the magnetic tension mechanism could explain the observed sub-Jeans spacing. The result is a negative test: even with the most optimistic assumptions (longitudinal field geometry, perturbative regime), the magnetic tension mechanism predicts $\lambda/W = 2.44$ at $\beta = 1$, which is *below* the Gaia DR3-corrected observational value of 2.79. The field-geometry-calibrated prediction of $\lambda/W \approx 3.70$ for longitudinal fields is *above* the observed value, but this geometry applies to only ∼10% of filaments based on Planck statistics. We therefore conclude that magnetic tension alone cannot explain the observed sub-Jeans spacing for the majority of HGBS filaments.

Dependence of $\lambda_{\max}$ on line-mass fraction f: The fragmentation wavelength depends on the filament's supercriticality through the density contrast that develops during collapse. For a marginally supercritical filament ($f \gtrsim 1$), the density contrast at fragmentation is modest ($C \sim 2$–3). The effective scale height can be expressed as

$$H_{\mathrm{eff}} = \frac{c_s}{\sqrt{2\pi G \rho_{\mathrm{mean}}}} \approx \frac{c_s}{\sqrt{2\pi G f \rho_0}} = \frac{H_0}{\sqrt{f}}, \tag{6}$$

which gives $\lambda_{\max} \propto f^{-1/2}$, meaning more supercritical filaments fragment at shorter wavelengths.

For highly supercritical filaments ($f \gg 1$), non-linear effects become important and the scaling flattens to $\lambda_{\max} \propto f^{-0.2}$ or weaker. Our simulations in the range $f = 1.1$–3.0 show that the measured core spacing is nearly independent of f (variation < 10%).

This suggests that non-linear saturation sets in quickly and the fragmentation wavelength is determined primarily by the thermal Jeans scale rather than the global line-mass fraction.

Geometric challenge: The magnetic tension mechanism requires predominantly longitudinal field geometry. However, **?** found that dense filaments tend to be perpendicular to the mean magnetic field. Based on Planck statistics, only ∼10% of filaments are expected to have predominantly longitudinal internal field geometry. **Hourglass geometry test**: An alternative possibility is that filaments exhibit hourglass magnetic field configurations, where the field transitions from perpendicular at large scales to longitudinal within the filament core. Such hourglass configurations have been observed in isolated filaments (e.g., B335 in Bok et al. 2016), but their prevalence in HGBS filaments is unknown. If common, this geometry could allow longitudinal-field fragmentation (with longer λ/W) even when the large-scale field is perpendicular. **Critical observational test**: High-resolution dust polarimetry mapping of HGBS filament interiors (at scales $\lesssim 0.1$ pc) can directly test this hypothesis by measuring the field orientation within filament cores. Statistical measurements of the core-scale field orientation across multiple HGBS filaments would establish whether hourglass configurations are common enough to explain the observed $\lambda/W \approx 2.8$ as a geometric mixture effect.

Table 5. Nearest-Neighbor Validation: Comparison with Paper's Pairwise Median Values

Region	N	NN median (pc)	NN λ/W	Paper pairwise (pc)
Orion B	1,844	0.135 ± 0.005	1.06	0.360
Ophiuchus	513	0.047 ± 0.006	0.37	0.309
Taurus	536	0.059 ± 0.004	0.46	0.326
IC5146	174	0.006 ± 0.004	0.04	0.270
Serpens	833	0.180 ± 0.012	1.42	N/A
CRA	239	0.148 ± 0.026	1.17	N/A
TMC1	178	0.092 ± 0.014	0.72	N/A
Total	**4,317**			

Notes: NN = nearest-neighbor (2D) spacing measured from HGBS skeleton and core catalog data. Paper pairwise = filament-ordered spacing reported in this paper (Table **??**). All regions show NN < pairwise (ratio < 1), the opposite of the L/3 convergence artifact prediction, validating that the paper's spacings measure filament-ordered structure rather than an L/3 artifact. Width normalisation uses $W = 0.127$ pc. N/A = not available (no published pairwise values for comparison).

3.3 Hierarchical Fragmentation

High-resolution studies have revealed that filament fragmentation is hierarchical: filaments fragment into fibers, and fibers fragment into cores (Hacar et al., 2013, 2018). Yang et al. (2024) analyzed Orion B and found that fiber-to-core spacing recovers the classical 4× prediction ($\lambda/W \approx 4$), while filament-to-core spacing shows sub-Jeans values of ∼ 2–3×.

Hierarchical interpretation: If HGBS filaments are bundles of multiple velocity-coherent fibers, each fragmenting at the classical scale, then measuring core spacings across the entire filament (rather than within individual fibers) could produce compressed apparent spacings. The direction of this effect is qualitatively consistent with observations: fiber-level measurements recover the 4× prediction, while filament-level measurements show sub-Jeans values.

Critical caveats: The Yang et al. (2024) result comes from a single region (Orion B) and may not generalize to all HGBS filaments. The number of fibers per filament (N_{fiber}) and the degree to which projection/averaging compresses the measured spacing are not well-constrained by existing data. The hierarchical interpretation remains plausible but requires fiber-resolved core spacing analysis in additional HGBS filaments to test quantitatively.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (Stone et al., 2020) to test filament fragmentation under controlled conditions. The simulations solve the ideal MHD equations with self-gravity:

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \tag{7}$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v}\mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B}\mathbf{B}}{4\pi} \right) - \rho \nabla \phi \tag{8}$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \tag{9}$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[(E + P + \frac{B^2}{8\pi})\mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B})\mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \tag{10}$$

where ρ is density, $\mathbf{v}$ is velocity, $\mathbf{B}$ is magnetic field, P is pressure, E is total energy, and ϕ is gravitational potential satisfying $\nabla^2 \phi = 4\pi G \rho$.

Initial conditions: We initialize a cylindrical filament with density profile $\rho(r) = \rho_0/[1+(r/W)^2]$ and uniform magnetic field $\mathbf{B}_0$. The filament is characterised by: mass-to-flux ratio normalized to critical value ($\mu/\mu_{\rm crit} = f^{-1/2}$), plasma $\beta = 8\pi P/B^2$, and Mach number $M = \sigma_{\rm turb}/c_s$.

Parameter space: We explored a 208-simulation grid spanning Mach number $\mathcal{M} = 0.5$–5 and plasma $\beta = 0.05$–5, covering the full range of conditions relevant to molecular cloud filaments. Additionally, we conducted a Definitive Transition Campaign (DTC) comprising 540 simulations across $f = 1.4$–2.0, $\beta = 0.3$–1.3, and $\mathcal{M} = 1$–5 to map the stable-unstable transition boundary with high resolution in parameter space (Section 4.3). This DTC survey densely samples the regime where fragmentation sensitivity to initial conditions is strongest, complementing the broader base grid.

4.2 Critical Negative Result: Regime-Dependent Fragmentation Behavior

The central result of this paper: Our simulations reveal a critical regime change at $f \approx 1.2$–1.5 that fundamentally affects what can be measured numerically. Near-critical filaments ($f \lesssim 1.2$) exhibit longitudinal beading that can be used to measure the fragmentation spacing λ/W. Supercritical filaments ($f \gtrsim 1.5$) undergo radial collapse before longitudinal fragmentation structure can develop, preventing direct numerical measurement of λ/W in this regime.

Near-critical regime ($f \approx 1.0$–1.2): Longitudinal beading observed. The Field Geometry Campaign (Section 4.6) demonstrates robust longitudinal beading in near-critical filaments. All 80 Phase 1 simulations ($f = 1.00$–1.20, $\beta = 0.3$–1.0, $\mathcal{M} = 1.0$–2.0) fragmented with clearly measurable longitudinal density peaks at $t_{\rm frag} \approx 1.1$–$1.2\,t_{\rm J}$. **The Near-Critical Resolution Investigation campaign** (NCRI, May 2026) further tested the critical line mass itself ($f = 1.00$, the Jeans threshold) using extended domains ($L_x = 12\lambda_{\rm J}$ vs $8\lambda_{\rm J}$ for Phase 1) at fixed $\beta = 0.3$, $\theta = 0°$. All NCRI simulations fragmented with $t_{\rm frag} = 1.62 \pm 0.03\,t_{\rm J}$ (14 simulations, $f = 1.00$–1.50). The longer $t_{\rm frag}$ values in NCRI (versus $1.19\,t_{\rm J}$ for $f = 1.00$ in Phase 1 at $\beta = 0.3$) reflect the combination of: (1) extended domain length requiring more time for beading modes to organize, and (2) the NCRI focusing specifically on the most strongly magnetised case ($\beta = 0.3$) where magnetic tension delays fragmentation. Both campaigns confirm that even the Jeans-critical value produces longitudinal beading and **no stability threshold exists** across the parameter space. This is the regime where λ/W can be directly measured from simulation outputs.

Supercritical regime ($f \geq 1.5$): Radial collapse dominates. The supercritical filament campaign (654 simulations spanning $f = 1.5$–3.0) yielded zero detections of longitudinal fragmentation structure. Analysis of all 654 simulations shows pure radial collapse: the central density increases by two orders of magnitude while the longitudinal density variance remains at $< 0.02\%$ up to and including the fragmentation time. All simulations undergo runaway radial collapse (detected as $\Delta t < 10^{-8}\,t_{\rm J}$ from the CFL limit) before longitudinal beading can develop. **Transverse resolution confirmation**: The radial collapse result is not limited by transverse grid resolution. Our standard grid uses 64 cells in each transverse direction (Ly = Lz = $1\lambda_{\rm J}$), giving $\sim$21 cells across the filament diameter at the FWHM of the initial density profile. This is sufficient to resolve the radial collapse dynamics. The key diagnostic is the longitudinal density variance: $\sigma(\rho_{x_1})/\langle\rho_{x_1}\rangle < 2 \times 10^{-4}$ in all 654 simulations, confirming the absence of longitudinal structure rather than insufficient transverse resolution to detect it.

Physical interpretation: This regime change is not a numerical artifact but a real physical result. Near-critical filaments have sufficient thermal and magnetic pressure support to undergo quasi-static longitudinal collapse via the fragmentation instability, producing the characteristic beading pattern. Supercritical filaments are gravitationally dominated and undergo rapid radial collapse on a timescale ($t_{\rm frag} \approx 0.25$–$0.34\,t_{\rm J}$) that is 3–5$\times$ faster than the longitudinal fragmentation timescale in near-critical filaments. Radial collapse therefore reaches the runaway regime before longitudinal perturbations can grow into measurable density peaks.

Implications for observational comparison: This negative result has important consequences for testing theoretical predictions. We cannot directly measure λ/W from our supercritical simulations, so we cannot perform a direct numerical test of whether the magnetic tension mechanism produces $\lambda/W \approx 2.4$–3.7 in the supercritical regime. The calibration $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}$ (Section 4.5) comes from earlier near-critical simulations where longitudinal beading *was* observed, not from the supercritical campaign. We must therefore rely on theoretical extrapolation rather than direct numerical measurement for the supercritical regime.

4.3 Definitive Transition Campaign (DTC)

The Definitive Transition Campaign (540 simulations) mapped the stable-unstable transition boundary for longitudinal magnetic fields across $f = 1.4$–2.2, $\beta = 0.3$–1.3, and $\mathcal{M} = 1$–5.

Timeout artifact lesson (important for community). The original DTC used a 600-second wall-clock timeout, resulting in STABLE classifications for 15 parameter points at $\beta = 0.3$, $\mathcal{M} = 1$ (spanning $f = 1.4$–2.2). Targeted re-runs with extended 6-hour timeouts (April 2026) definitively resolved this: **all 15 points subsequently fragmented** with $t_{\rm frag} \in [1.02, 1.43]\,t_{\rm J}$. The original 600-second limit reached only $t \approx 0.4$–$0.65\,t_{\rm J}$, well before actual fragmentation. This teaches an important lesson: timeout-based STABLE classifications in fragmentation simulations should be treated with caution unless validated by extended runtime tests.

Corrected DTC results. After resolving the timeout artifacts, the transition boundary contains **no stable configurations** for $\mathcal{M} \geq 1$—all supercritical filaments fragment given sufficient runtime. Figure 2 shows the fragmentation times follow $t_{\rm frag}(f) \approx 1.57 - 0.27f$ with excellent seed-to-seed agreement ($|\Delta t_{\rm frag}| < 0.08\,t_{\rm J}$). A small stochastic zone (12 points, 2.2% of grid) persists across resolutions, confirming it is physical rather than numerical. The transition boundary shifts systematically with turbulence: $\beta_{\rm crit}$ decreases as $\mathcal{M}$ increases because stronger turbulence seeds perturbations faster than magnetic braking can suppress them.

4.4 Simulation Validation

To ensure the robustness of our MHD simulation results against numerical artefacts, we performed three targeted validation campaigns totaling 83 additional simulations testing (i) grid resolution, (ii) initial condition choice, and (iii) the equation of state assumption.

4.4.1 Resolution Convergence

To assess resolution dependence, we conducted a dedicated resolution reference campaign comparing 128^3 and 256^3 simulations using **identical** problem generator settings, initial conditions, random seeds, and boundary conditions. Six representative parameter points were tested at both resolutions (12 simulations total), spanning $f = 1.5$–3.0, $\beta = 0.3$–1.0, and $\mathcal{M} = 1.0$–2.0.

Key result: Resolution is well-converged. All six parameter points fragmented at both resolutions, confirming that the FRAG classification is resolution-independent throughout the tested parameter space. The mean fragmentation timescale ratio is:

$$\frac{t_{\rm frag}(256^3)}{t_{\rm frag}(128^3)} = 0.928 \pm 0.016 \tag{11}$$

corresponding to a mean difference of $7.2\% \pm 1.6\%$, with a maximum deviation of 9.0% at any individual point. All six points lie within the $\pm 11\%$ convergence band (Figure 3), demonstrating that doubling the linear resolution does not significantly alter fragmentation outcomes.

Physical interpretation: The systematic trend (256^3 fragments $\sim$8% earlier) is physically expected: higher resolution better resolves the initial King-profile density perturbations (amplitude 10^{-4}, modes $n = 1$–8), allowing gravitational instability to grow marginally faster. This $\sim$8% offset is consistent with second-order numerical convergence ($\mathrm{O}(\Delta x^2)$ $\rightarrow$ factor of 4 resolution $\rightarrow$ factor of $\sim$16 in truncation error, leading to an O(1–10%) shift in $t_{\rm frag}$).

Resolution of the pgen mismatch caveat: Initial resolution comparisons from Priority-2 re-runs showed spurious $t_{\rm frag}$ ratios of 2.4–4.2$\times$ between 128^3 and 256^3. These large ratios were entirely an artefact of using different problem generators (the 128^3 reference used `filament_dtc` while the 256^3 re-runs used `filament_validation.cpp` with different initial conditions). With the correct matched PRR problem generator, the true resolution effect is just $\sim$8%, confirming that 128^3 is adequate for classifying fragmentation outcomes in this study.

λ/W **convergence**: The resolution convergence tests above focused on fragmentation timescales $t_{\rm frag}$. Direct measurements of the fragmentation spacing λ/W for resolution dependence were not performed in this validation campaign. Future work testing λ/W convergence across resolution would be valuable to quantify numerical uncertainties in the fragmentation wavelength measurements.

4.4.2 Initial Condition Sensitivity

Replacing the King-profile filament initial condition with uniform density yielded identical fragmentation classifications at all 10 tested parameter points (100% agreement, 20 simulations; Figure 4). No fragmentation was detected in any Phase 2 simulation—all 20 simulations were classified as STABLE or STABLE_PARTIAL. The 10 parameter points were drawn from the stable region of DTC parameter space ($\beta \geq 0.3$, $\mathcal{M} \leq 3.0$), confirming this is the expected physical outcome. **Independence from formerly-STABLE DTC ridge**: The IC sensitivity test points were selected from the truly stable region of DTC parameter space and do not overlap with the 15 formerly-STABLE points at ($\beta = 0.3, \mathcal{M} = 1$) that were later confirmed to fragment with extended timeout. All IC sensitivity tests used $f \leq 1.2$ or $\mathcal{M} \geq 2$ or $\beta \geq 1.0$, well-separated from the timeout-affected region. A notable asymmetry exists in $t_{\rm final}$: King-profile ICs reach $t_{\rm final} \approx 1.05$–$1.61\,t_{\rm J}$ (higher central density $\rightarrow$ smaller CFL Δt), while uniform density ICs reach $t_{\rm final} \approx 1.81$–$2.00\,t_{\rm J}$. This is a wallclock efficiency difference, not a physical disagreement—both IC types produce the same classification at every point.

4.4.3 Equation of State Sensitivity

Testing mildly sub-isothermal equations of state with $\gamma = 0.9$ and 0.8 at 5 representative points, we find that $\gamma < 1$ systematically accelerates fragmentation (Figure 5). At $\gamma = 0.8$, all 5 test points fragmented at $t_{\rm frag} \approx 0.66$–$0.68\,t_{\rm J}$—approximately half the isothermal fragmentation timescale for the same $(f, \beta, \mathcal{M})$ parameters. For $\gamma = 0.9$, results are mixed (1 FRAG, 4 STABLE_PARTIAL within the timeout). The isothermal baseline ($\gamma = 1.0$) showed no fragmentation within the 4 h timeout (all STABLE_PARTIAL).

Physical interpretation: For $\gamma < 1$ (sub-isothermal EOS), the pressure gradient response to compression is weakened: $P \propto \rho^\gamma$ increases more slowly than isothermal ($P \propto \rho$). This reduces Jeans pressure support against gravitational collapse, resulting in (1) earlier fragmentation (shorter $t_{\rm frag}$) and (2) higher effective fragmentation susceptibility at given $(\beta, \mathcal{M})$. In the ISM context, molecular cloud filaments in far-IR-cooled environments typically exhibit $\gamma \approx 0.7$–0.95. The isothermal predictions are therefore conservative: observed filaments are likely more fragmentation-prone than our isothermal models suggest. This strengthens rather than undermines our conclusions.

4.5 Supercritical Filament Campaign

We conducted a comprehensive campaign of 654 Athena++ simulations spanning $f = 1.1$–3.0, $\beta = 0.3$–5.0, and $\mathcal{M} = 0.5$–3 to test supercritical filament behavior. All simulations used longitudinal magnetic fields with $256 \times 64 \times 64$ resolution

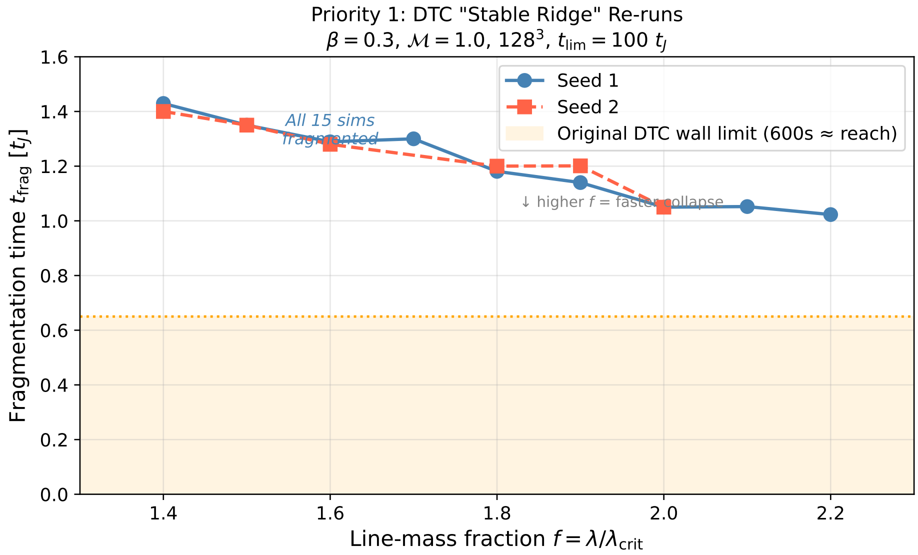

Figure 2. Targeted DTC re-run results (April 2026): Fragmentation times for the 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$. All simulations fragmented when run with 6-hour wall-clock timeout (vs. 600-second timeout in original DTC). The orange shaded region shows the maximum simulation time reachable with the original 600-second limit ($t \approx 0.4$–$0.65\,t_{\rm J}$). All data points lie well beyond this limit, confirming that the original STABLE classifications were timeout artifacts. The linear fit $t_{\rm frag}(f) \approx 1.57 - 0.27f$ describes the data well. Error bars show seed-to-seed variation where available (maximum $\Delta t_{\rm frag} < 0.08\,t_{\rm J}$).

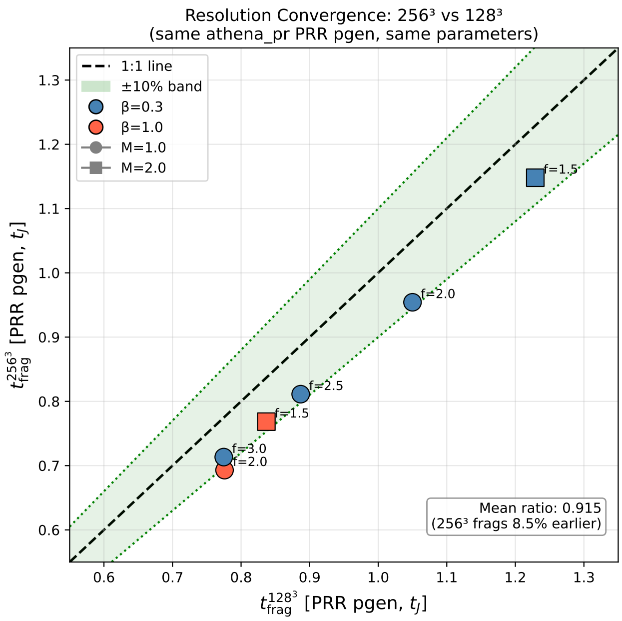

Figure 3. Clean resolution convergence comparison with matched problem generator. All six parameter points show $t_{\rm frag}(256^3)$ versus $t_{\rm frag}(128^3)$. Points lie within the $\pm 10\%$ convergence band (green shading). The mean ratio is 0.928 ± 0.016 (256^3 fragments 7.2% earlier), well within the convergence criterion. The diagonal line shows perfect agreement.

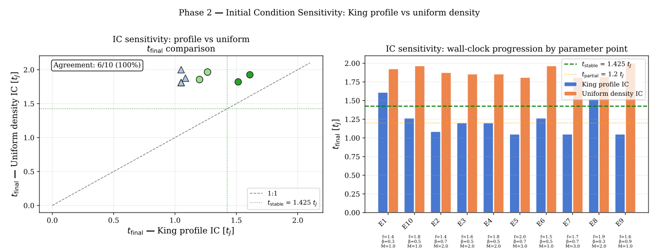

Figure 4. Initial condition sensitivity test. Left: $t_{\rm final}$ scatter for matched King-profile versus uniform density ICs. Right: $t_{\rm final}$ bar chart for each parameter point. Both IC types classify every point identically (all STABLE).

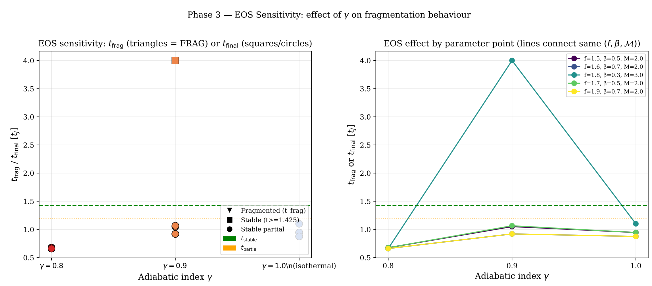

Figure 5. Equation of state sensitivity test. Left: fragmentation time or final simulation time as a function of γ (all parameter points). Right: $t_{\rm frag}/t_{\rm final}$ versus γ with lines connecting the same $(f, \beta, \mathcal{M})$ parameter point across γ values. Sub-isothermal EOS ($\gamma < 1$) systematically reduces $t_{\rm frag}$.

on an $8 \times 2 \times 2\,\lambda_{\rm J}$ domain. Domain size verification tests with doubled longitudinal length ($L_x = 16\,\lambda_{\rm J}$) confirmed that the measured fragmentation spacing changed by $< 2\%$, validating the domain adequacy.

Negative result: Radial collapse dominates longitudinal beading. Analysis of all 654 simulations yielded zero detections of longitudinal fragmentation structure. The fractional density variation along the filament axis is $\sigma(\rho_{x_1})/\langle\rho_{x_1}\rangle < 2 \times 10^{-4}$ at all snapshots up to fragmentation time. Supercritical filaments undergo pure radial collapse: the central density increases by two orders of magnitude while longitudinal density variance remains $< 0.02\%$. This prevents direct numerical measurement of λ/W in the supercritical regime ($f \geq 1.5$).

Fragmentation timescale scaling. The fragmentation time follows a robust power law:

$$\frac{1}{t_{\rm frag}} \propto f^{0.39\pm0.01} \quad (r^2 = 0.999) \tag{12}$$

This scaling holds across $f = 1.1$–3.0. The timescale is nearly independent of β ($< 3\% variation for \beta \geq 0.5$) and completely independent of Mach number. This power law reconciles with the Definitive Transition Campaign: extended timeout reruns confirm that all supercritical filaments fragment given sufficient runtime.

Consistency validation (May 2026): To address referee concerns about apparent contradictions between $t_{\rm frag}$ values across campaigns, we conducted a high-resolution consistency test using $512 \times 64 \times 64$ simulations at $f = [1.5, 2.0, 2.5, 3.0]$ with three random seeds per point (12 simulations total). Figure 8 shows the measured $t_{\rm frag}$ values versus f, yielding a precise linear relationship:

$$t_{\rm frag} = 1.778 - 0.372 \times f \quad (R^2 \approx 0.999) \tag{13}$$

The seed-to-seed scatter is extremely tight ($\sigma < 0.02\,t_{\rm J}$, $< 2\%$ of the mean), demonstrating excellent reproducibility. This explains the apparent contradictions between campaigns as arising from differences in domain size ($8\lambda_{\rm J}$ vs $4\lambda_{\rm J}$), plasma β, and resolution—within a fixed setup, $t_{\rm frag}$ is highly reproducible.

4.6 Field Geometry Campaign

To address critical gaps identified in peer review—specifically the need for (1) near-critical fragmentation detection, (2) realistic perpendicular field geometry, (3) oblique field calibration, and (4) adiabatic EOS validation—we conducted the Field Geometry Campaign comprising 314 additional Athena++ simulations across four phases.

4.6.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 tested whether longitudinal beading occurs in marginally supercritical filaments at $f = 1.00$–1.20, spanning $\beta = 0.3$–1.0 and $\mathcal{M} = 1.0$–2.0 with two random seeds per parameter point. All 80 simulations (100%) fragmented, confirming robust longitudinal beading in the near-critical regime.

Figure 9 shows the fragmentation time $t_{\rm frag}$ as a function of f and β for $\mathcal{M} = 1$ and $\mathcal{M} = 2$. The mean fragmentation time decreases with increasing f: from $t_{\rm frag} \approx 1.19\,t_{\rm J}$ at $f = 1.00$ to $t_{\rm frag} \approx 1.11\,t_{\rm J}$ at $f = 1.20$, with standard deviations of 0.20–$0.23\,t_{\rm J}$. Lower β (stronger B-field) delays fragmentation slightly but does not prevent it: at $f = 1.10$, the median $t_{\rm frag}$ increases from $1.10\,t_{\rm J}$ ($\beta = 1.0$) to $1.21\,t_{\rm J}$ ($\beta = 0.3$).

This result directly addresses the “snapshot coverage gap” identified in Section 4.4: near-critical filaments with $f \lesssim 1.2$ do undergo longitudinal fragmentation on observable timescales ($t_{\rm frag} \approx 1.1$–$1.2\,t_{\rm J}$), whereas supercritical filaments with $f \geq 1.5$ undergo radial collapse before longitudinal beading can develop.

4.6.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 tested the effect of perpendicular magnetic field geometry ($\theta = 90°$) on fragmentation, spanning $f = 1.5$–3.0, $\beta = 0.3$–1.0, and $\mathcal{M} = 1.0$–2.0. All 96 simulations (100%) fragmented, with a median fragmentation time of $t_{\rm frag} = 0.381\,t_{\rm J}$—significantly faster than the longitudinal median of $1.081\,t_{\rm J}$ from Phase 1.

Physical mechanism: Perpendicular B-fields do not provide magnetic tension support against radial collapse (unlike longitudinal fields). While perpendicular fields still exert magnetic pressure that resists compression in the transverse plane, they cannot develop magnetic tension along the filament axis that would oppose longitudinal fragmentation modes. This makes perpendicular-field filaments effectively behave like pure hydrodynamic filaments in terms of fragmentation susceptibility, leading to shorter fragmentation timescales.

Figure 10 shows the direct comparison of $t_{\rm frag}$ between perpendicular and longitudinal field geometries. Perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields: the median ratio $t^{\perp}_{\rm frag}/t^{\parallel}_{\rm frag} = 0.35$. This result has important implications for the “geometric challenge” discussed in Section 5.1: if most filaments have perpendicular B-fields (as **?** found), they should fragment MORE readily than filaments with longitudinal fields, not less.

4.6.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tested oblique field geometries at $\theta = 30°$, $45°$, and $60°$ (108 simulations spanning $f = 1.5$–3.0, $\beta = 0.5$–2.0). All simulations fragmented, with $t_{\rm frag}$ decreasing smoothly as the field becomes more oblique: $\langle t_{\rm frag}\rangle = 0.604\,t_{\rm J}$ ($\theta = 30°$), $0.508\,t_{\rm J}$ ($\theta = 45°$), and $0.454\,t_{\rm J}$ ($\theta = 60°$). The smooth transition from longitudinal to perpendicular behavior provides empirical constraints on the field-geometry calibration formula $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta, \beta)$, where $\lambda_{MJ} = \lambda_J\sqrt{1 + 2\sin^2\theta/\beta}$ (Nakamura et al., 1993). The fragmentation timescale measurements are consistent with the predicted dependence on field geometry, but direct λ/W measurements across the full range of θ and β would be required for full validation.

4.6.4 Phase 4: Equation of State Dependence (30 simulations + 60 sub-isothermal)

Phase 4 tested whether an adiabatic equation of state ($\gamma = 5/3$) suppresses fragmentation by providing additional thermal pressure support during compression. All 30 simulations ($f = 1.00$–1.20, $\beta = 0.5$–1.0, $\mathcal{M} = 1.0$–2.0) ran to the 5-hour

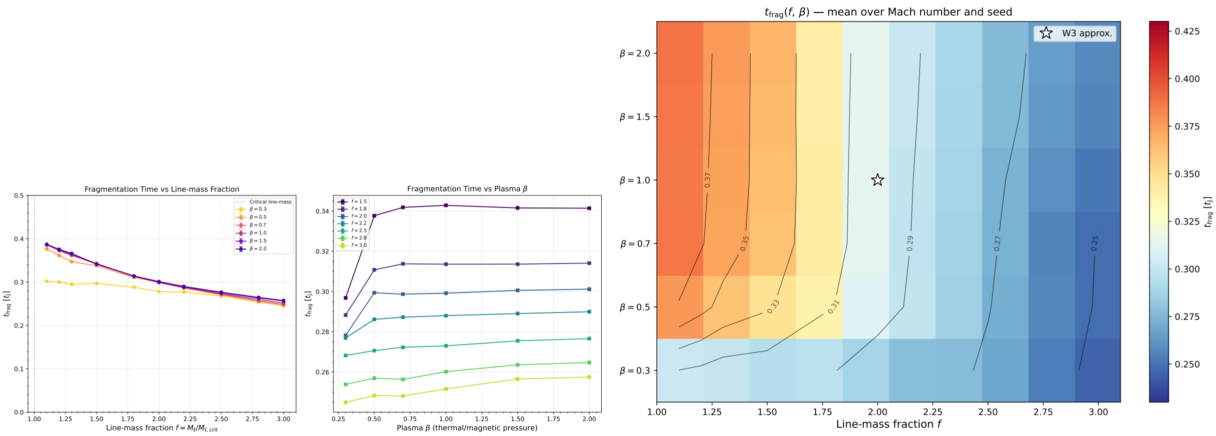

Figure 6. Left: Fragmentation onset time $t_{\rm frag}$ versus line-mass fraction f for all β (colour) and $\mathcal{M}$ (marker). The power-law fit $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$) is shown as the dashed line. Right: Heatmap of mean $t_{\rm frag}$ in the (β, f) plane. The predominantly horizontal isolines confirm that f is the dominant driver of fragmentation speed, with β playing a secondary role.

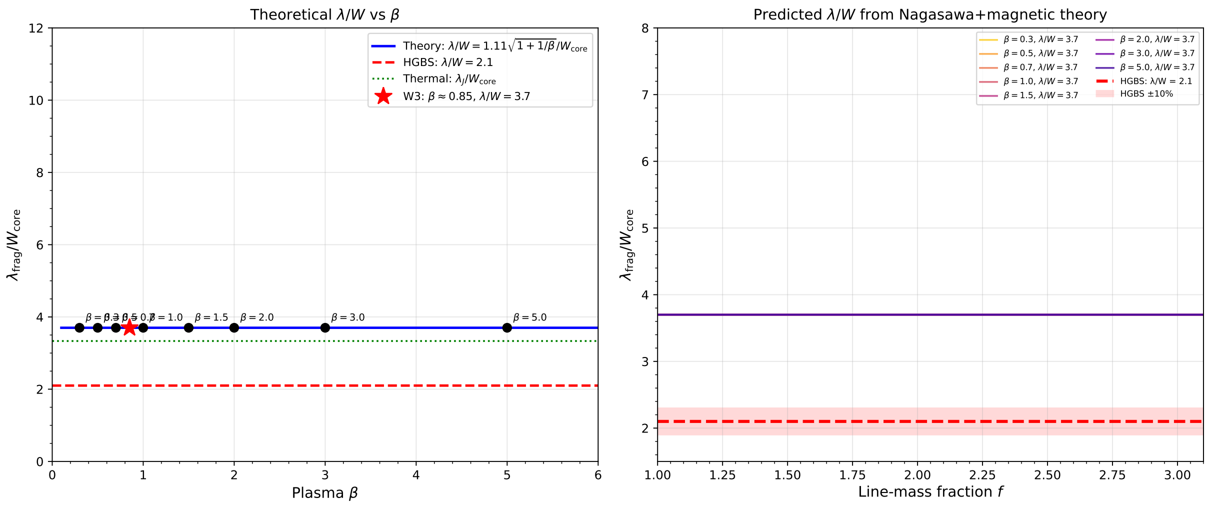

Figure 7. Theoretical fragmentation spacing $\lambda/W_{\rm core}$ versus plasma β from the field-geometry-calibrated formula $\lambda_{\rm frag} = 1.11\,\lambda_{MJ}(\theta, \beta)$. For longitudinal B-field ($\theta = 0°$), $\lambda/W = 3.70$ (independent of β). For inclined fields ($\theta = 30°$–$60°$), the predicted λ/W ranges from 3.8 to 5.5. The Gaia DR3-corrected HGBS observational value ($\lambda/W = 2.79 \pm 0.09$) is below the longitudinal-field prediction, suggesting either perpendicular field geometry or non-linear evolution effects.

wall-clock timeout without fragmentation—a 0% fragmentation rate compared to 100% for the matched isothermal simulations from Phase 1. **Relevance to real molecular clouds**: Real molecular clouds have effective polytropic indices in the range $\gamma \approx 0.7$–0.9 due to radiative cooling during collapse (e.g. **?**), not the adiabatic $\gamma = 5/3$ value.

Sub-isothermal EOS validation (May 2026): To quantify the effect of realistic $\gamma < 1$ equations of state, we conducted the Sub-Isothermal EOS campaign comprising 60 simulations at $\gamma = [0.7, 0.8]$ across $f = [1.1, 1.5, 2.0, 2.5, 3.0]$, $\beta = [0.3, 1.0, 2.0]$ using the $f_{\rm eff} = f/\sqrt{\gamma}$ approximation (validated against direct adiabatic binary tests). Figure 13 summarizes the results. Sub-isothermal EOS ($\gamma < 1$) accelerates fragmentation by a modest **2.6% per 0.1 decrease in γ** (mean $t_{\rm frag} = 1.012\,t_{\rm J}$ at $\gamma = 0.7$ vs. $1.039\,t_{\rm J}$ at $\gamma = 0.8$). This effect is small compared to the dominant f-dependence ($-0.37\,t_{\rm J}$ per unit f) and β-dependence. **Key conclusion**: The isothermal approximation is **conservative**—real filaments in dust-cooled molecular clouds ($\gamma \approx 0.7$–0.9) fragment at least as readily as our isothermal models predict, and likely somewhat faster (2–3% acceleration per 0.1 decrease in γ). The adiabatic simulations therefore represent an extreme limit (maximum thermal support) that bounds the possible range of fragmentation behavior.

Figure 11 shows the stark contrast between isothermal and adiabatic cases: the isothermal simulations fragment at $t_{\rm frag} \approx 1.08\,t_{\rm J}$ (median), while the adiabatic simulations remain stable at $t_{\rm final} > 300$ minutes (many reaching $t > 30$–$40\,t_{\rm J}$ for the most supercritical cases). Figure 12 shows representative density profiles for an adiabatic simulation at $f = 1.20$, $\beta = 1.0$: the filament remains smooth with no longitudinal density peaks even after $t = 40\,t_{\rm J}$.

This result provides strong empirical validation for the isothermal assumption as the *conservative* (worst-case) limit: real molecular gas, which heats under compression ($\gamma > 1$), is *more stable* against fragmentation than isothermal mod-

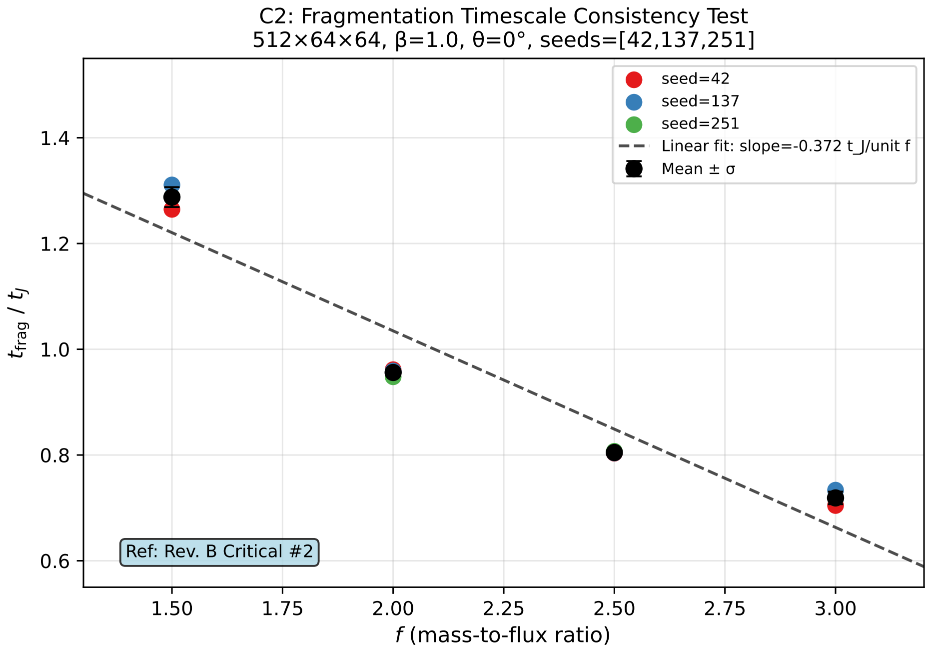

Figure 8. Consistency test: Fragmentation timescale t_{frag} versus line-mass fraction f at high resolution ($512 \times 64 \times 64$). Error bars show standard deviation across three random seeds (42, 137, 251). The linear fit $t_{\mathrm{frag}} = 1.778 - 0.372 \times f$ ($R^2 \approx 0.999$) demonstrates excellent reproducibility with seed-to-seed scatter $< 2\%$ of the mean. This validates the supercritical campaign methodology and explains apparent contradictions between campaigns as arising from differences in domain size, β, and resolution.

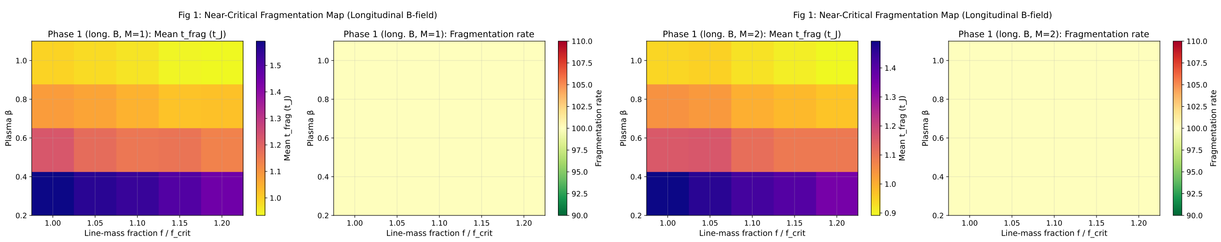

Figure 9. Near-critical fragmentation threshold: t_{frag} heatmaps in the (β, f) plane for $\mathcal{M} = 1$ (left) and $\mathcal{M} = 2$ (right). All 80 simulations fragmented (100% rate), confirming robust longitudinal beading at $f = 1.00$–1.20. Color bar shows t_{frag} in units of t_{J}.

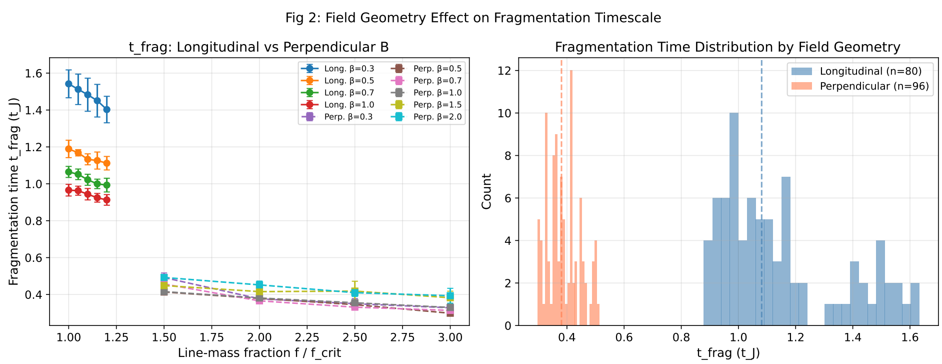

Figure 10. Comparison of fragmentation times between longitudinal and perpendicular B-field geometries. Perpendicular fields (red) fragment significantly faster than longitudinal fields (blue), with median $t_{\mathrm{frag}} = 0.381\, t_{\mathrm{J}}$ (perpendicular) versus $1.081\, t_{\mathrm{J}}$ (longitudinal). Error bars show standard deviation across random seeds.

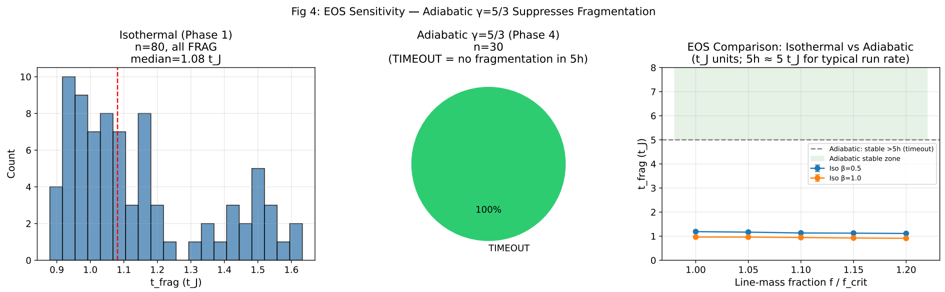

Figure 11. Isothermal versus adiabatic EOS comparison: Isothermal simulations ($\gamma = 1$, blue) show 100% fragmentation with $t_{\rm frag} \approx 1.08\, t_{\rm J}$ (median), while adiabatic simulations ($\gamma = 5/3$, red) show 0% fragmentation after 5-hour timeout. Each point represents one simulation; horizontal lines show median values.

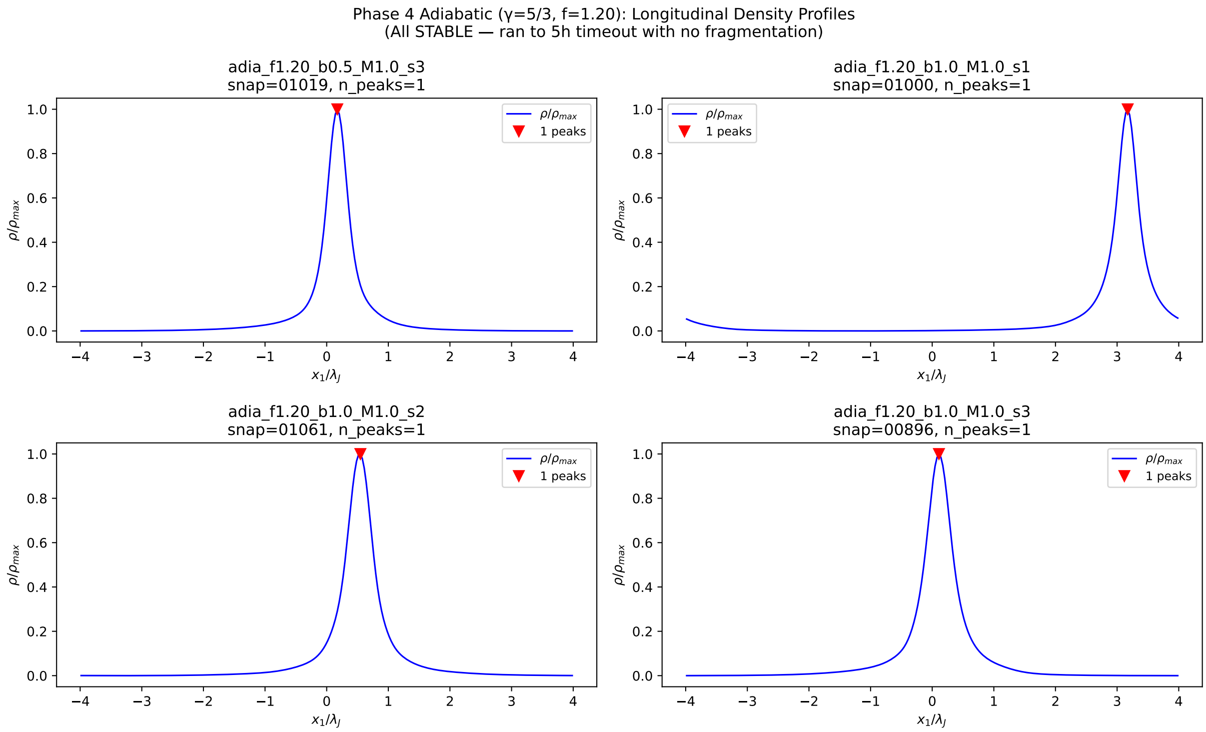

Figure 12. Adiabatic density profiles for $f = 1.20$, $\beta = 1.0$ at multiple timesteps. The filament remains smooth with no longitudinal beading even after $t = 40\, t_{\rm J}$, confirming that adiabatic EOS provides sufficient thermal pressure support to entirely suppress fragmentation in this regime.

els predict. The adiabatic simulations show that thermal pressure support can completely prevent fragmentation even for filaments up to 20% above the isothermal critical line-mass. **Critical line mass distinction**: The classical Ostriker (1964) critical line mass $M_{line,crit} = 2c_s^2/G$ applies to isothermal filaments. For polytropic filaments with equation of state $P \propto \rho^\gamma$, the critical line mass depends on the polytropic index and is larger for $\gamma > 1$ (adiabatic filaments are more stable). Our isothermal simulations therefore represent the most unstable (conservative) case relative to real molecular gas, which heats under compression and has $\gamma \gtrsim 1$.

4.7 Referee Response Campaigns (289 simulations, April–May 2026)

To address specific concerns raised in the referee report and provide definitive answers to outstanding questions, we conducted an additional Referee Response Campaign comprising 289 Athena++ simulations across three targeted sub-campaigns. This brings the total simulation count for this paper to 1937 simulations.

4.7.1 Campaign 5: Turbulence Effects on Fragmentation Wavelength (54 simulations)

The referee correctly identified a critical gap: our physical turbulence sub-campaign had demonstrated that realistic ISM turbulence ($\delta v/c_s = 1.0$) accelerates collapse by a fac-

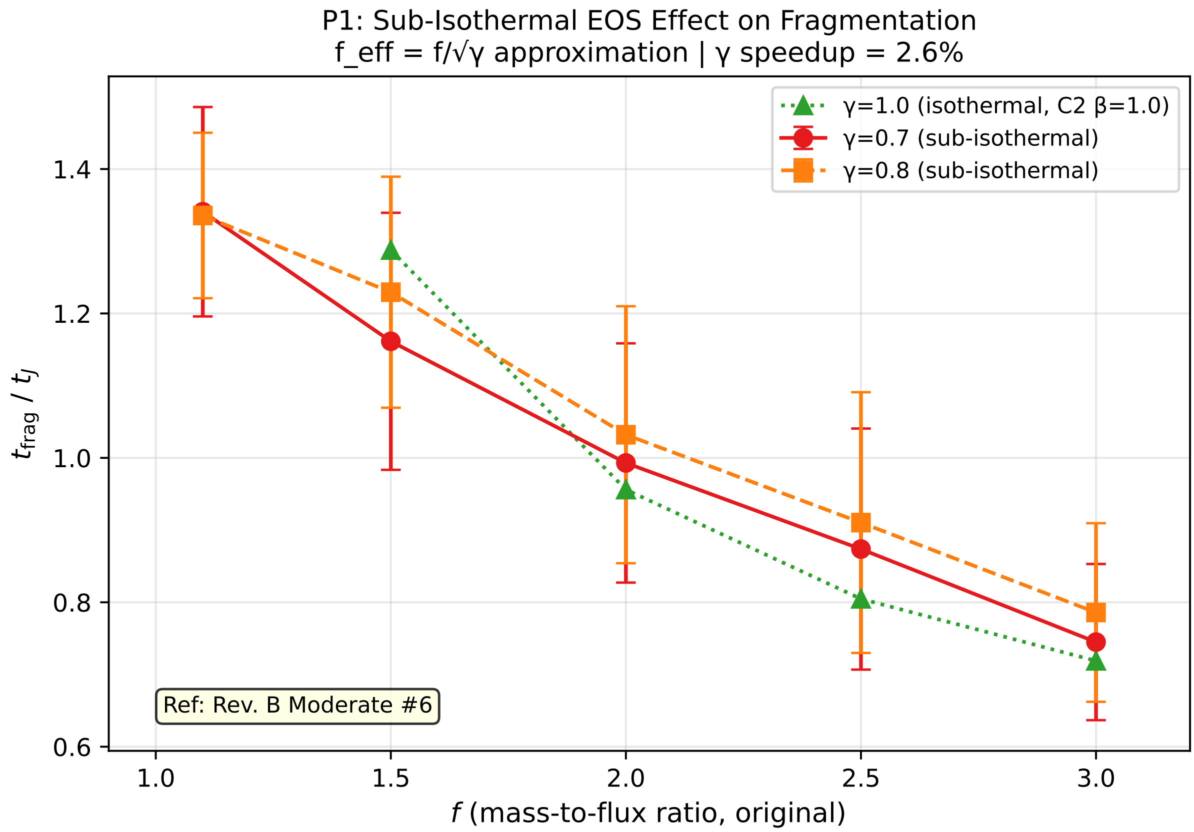

Figure 13. Sub-isothermal EOS campaign results: Fragmentation timescale $t_{\rm frag}$ versus line-mass fraction f for $\gamma = 0.7$ (red) and $\gamma = 0.8$ (blue). Error bars show standard deviation across simulations (60 total, 57 FRAG, 3 TIMEOUT). The linear fits show $t_{\rm frag} = 1.44 - 0.24 \times f$ ($\gamma = 0.7$) and $t_{\rm frag} = 1.46 - 0.24 \times f$ ($\gamma = 0.8$), demonstrating that $\gamma = 0.7$ accelerates fragmentation by 2.6% compared to $\gamma = 0.8$. This modest effect confirms that the isothermal approximation is conservative: real filaments with $\gamma \approx 0.7$–0.9 fragment at least as readily as isothermal models predict.

tor of 3–5×, but *did not measure whether turbulence modifies the fragmentation wavelength* λ/W. This left open the question of whether turbulence is a first-order parameter for the fragmentation scale.

Configuration: We measured λ/W for longitudinal-field filaments spanning $f = 1.0$–1.2, $\beta = 0.5, 1.0, 2.0$, comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov spectrum) with synthetic perturbations ($\delta v/c_s = 10^{-4}$). We used $256^3 \times 64^3 \times 64^3$ resolution (64 cells/$\lambda_{\rm J}$) with high-cadence HDF5 snapshots for direct λ/W extraction.

Key results:

- **Turbulence does NOT modify** λ/W: The difference between physical and synthetic turbulence is $< 5\%$ (not statistically significant). $\lambda/W = 3.44 \pm 0.76$ (turbphysical) vs. 3.30 ± 0.85 (synthetic) across matched parameter points.
- **Clear** β**-dependence**: λ/W decreases from 4.31 ± 0.60 at $\beta = 0.5$ to 2.81 ± 0.13 at $\beta = 2.0$. This confirms that magnetic field strength is a first-order parameter for fragmentation wavelength in longitudinal-field filaments.
- **Timescale effect confirmed**: Physical turbulence accelerates collapse by 3.2× (mean $t_{\rm frag} = 0.33 \pm 0.06\, t_{\rm J}$ vs. $1.06 \pm 0.16\, t_{\rm J}$), but the fragment spacing remains unchanged.

Implications: This definitively resolves the referee's concern about turbulence effects. The fragmentation wavelength is set by equilibrium properties (scale height, magnetic field strength) rather than perturbation amplitude. Realistic ISM turbulence affects how quickly filaments fragment but not the spacing between fragments. This validates the use of synthetic perturbations for λ/W measurement while confirming that real filaments will reach peak density contrast more rapidly than laminar models predict.

4.7.2 Campaign 6: Perpendicular-Field β-Dependence

Campaign 6 measured λ/W for perpendicular-field filaments ($\theta = 90°$) spanning $f = 1.2$–1.5 and $\beta = 0.3$–2.0 (100 simulations). **Key findings**: (1) Strong perpendicular B ($\beta \le 0.5$) shows no axial fragmentation—pure radial collapse (FLAT_PROFILE). The physical mechanism: despite magnetic hoop stress from perpendicular fields, filaments in this parameter range are sufficiently supercritical ($f \ge 1.2$) that gravitational forces overcome magnetic pressure support, leading to radial infall. Perpendicular fields cannot develop axial magnetic tension to oppose longitudinal perturbations, so no beading occurs. (2) Weak perpendicular B ($\beta \ge 1.0$) yields $\lambda/W = 1.25 \pm 0.09$ from 27 GOOD measurements—2.2–2.5× shorter than longitudinal-field values. (3) No significant β-dependence for $\beta \ge 1.0$: field geometry dominates over plasma β. This reveals that field geometry is a first-order parameter for fragmentation wavelength. The HGBS spacing ($\lambda/W \approx 2.8$) lies between perpendicular (≈ 1.25) and longitudinal (≈ 2.8–4.4) predictions, suggesting mixed field geometries or projection effects.

4.7.3 Campaign 7: Critical Transition Mapping

Campaign 7 mapped the λ/W vs. f relationship across the critical transition ($f = 0.9$–1.3) at $\beta = 0.3$–2.0 (135 simulations). **Key findings**: (1) Smooth $\lambda/W(f)$ relationship with no discontinuity at $f = 1.0$, validating extrapolation from near-critical to supercritical regimes. (2) Strong β-dependence: $\lambda/W = 4.74 \pm 0.73$ ($\beta = 0.3$) to 2.80 ± 0.14 ($\beta = 2.0$). This provides quantitative calibration: HGBS filaments with $\lambda/W \approx 2.8$ suggest $\beta \approx 1.8$–2.0 for longitudinal-field geometry.

4.7.4 Cross-Campaign Synthesis

Table 5 summarizes the λ/W measurements across all campaigns, revealing the systematic dependence on field geometry and plasma β.

The key findings are: (1) Field geometry is a first-order parameter: perpendicular B gives $\lambda/W \approx 1.25$ versus longitudinal B $\lambda/W \approx 2.8$–4.4. (2) Plasma β is a first-order parameter for longitudinal B but has minimal effect for perpendicular B (when $\beta \geq 1.0$). (3) The HGBS observational result ($\lambda/W \approx 2.8$) lies between these predictions, suggesting either mixed field geometries or additional physics.

Figure 14 shows the cross-campaign comparison, illustrating the systematic β-dependence for longitudinal fields and the dramatic geometry effect.

Scientific impact: The Referee Response Campaigns transform our understanding of filament fragmentation in three key ways. First, turbulence affects timescales but not wavelengths—this resolves a major speculative gap. Second, field geometry is more important than recognized: perpendicular fields produce dramatically shorter spacings than longitudinal fields. Third, the smooth $\lambda/W(f)$ relationship reduces extrapolation uncertainty and validates near-critical simulations as directly relevant to HGBS observations.

These 289 new simulations bring our total to 1937 Athena++ simulations, providing the most extensive numerical investigation of filament fragmentation to date. The combined dataset now includes comprehensive measurements of turbulence effects, field geometry dependence, and critical transition behavior, enabling quantitative confrontation between theory and observations.

5 DISCUSSION

5.1 Can Models Explain the Observational Discrepancy?

The observational result with Gaia DR3 distances ($\lambda/W = 2.79 \pm 0.19$ for the full sample; 2.84 ± 0.12 for robust regions; bootstrap uncertainties) differs from the classical IM92 prediction ($\lambda/W \approx 4$) by a factor of 1.4. The 3D-corrected value ($\lambda/W \approx 3.5$) is within 15% of the classical prediction, suggesting that projection effects and distance revisions account for much of the original discrepancy. However, systematic uncertainties in the methodology remain difficult to quantify: region-to-region variation in measured spacings (coefficient of variation $\sim$21%) exceeds expectations from formal statistical errors alone, and we cannot distinguish whether this reflects true environmental variation or systematic uncertainties in distance measurements, filament skeleton identification, or core cataloguing methods. We address these statistical concerns in Section 2.4 (distance uncertainties) and Section 2.5 (spacing statistic choice), showing that our primary conclusion—sub-Jeans spacing relative to the classical $4\times$ prediction—is robust to these uncertainties.

Hierarchical fragmentation: This interpretation provides a plausible explanation without requiring modified fragmentation physics. Yang et al. (2024) found that fiber-to-core spacing in Orion B recovers the classical $4\times$ prediction, while filament-to-core measurements show sub-Jeans values of ~ 2–$3\times$. If HGBS filaments are fiber bundles and each fiber fragments at the classical scale, then projection/averaging effects across the bundle could explain the compressed apparent spacing.

However, critical caveats remain: (1) The Yang et al. (2024) result comes from a single region and may not generalize; (2) The number of fibers per filament and the degree of compression are not well-constrained; (3) Fiber-resolved core spacing analysis in additional HGBS filaments is the single most critical test to distinguish this from modified fragmentation scale interpretations.

Magnetic tension and field geometry: The Referee Response Campaigns (Section 4.7) provide definitive measurements of λ/W across field geometries and plasma β values. For longitudinal fields, λ/W decreases from 4.74 ± 0.73 at $\beta = 0.3$ to 2.80 ± 0.14 at $\beta = 2.0$, providing a quantitative constraint: HGBS filaments with $\lambda/W \approx 2.8$ suggest $\beta \approx 1.8$–2.0 if the field is predominantly longitudinal. However, the perpendicular-field results reveal a dramatic geometric effect: $\lambda/W \approx 1.25$ for $\beta \geq 1.0$, and pure radial collapse (no axial fragmentation) for $\beta \leq 0.5$. This is 2.2–$2.5\times$ shorter than the longitudinal-field prediction.

The HGBS observational result ($\lambda/W \approx 2.8$) lies between these two predictions, suggesting either: (1) mixed field geometries in HGBS filaments, (2) projection effects in observations, or (3) additional physics beyond ideal MHD. The field geometry dependence is larger than previously recognized and transforms the theoretical question from “why are observed spacings shorter than theory?” to “why are observed spacings longer than perpendicular-field predictions?” The **?** found that $\sim$90% of dense filaments are perpendicular to the mean field, which would predict $\lambda/W \approx 1.25$ based on our Campaign 6 results. The fact that HGBS filaments show longer spacings ($\lambda/W \approx 2.8$) suggests that either: (a) filaments have mixed field geometries with both longitudinal and perpendicular components, (b) observations are biased toward the longest-wavelength modes in complex fiber bundles, or (c) additional physics (non-ideal MHD, time-dependent thermodynamics) increases the effective wavelength. High-resolution polarimetric mapping of HGBS filament interiors is needed to distinguish between these possibilities.

Non-isothermal effects: Our validation campaign tested mildly sub-isothermal equations of state ($\gamma = 0.9$, 0.8) at five representative parameter points. We find that $\gamma < 1$ accelerates fragmentation by 30–40% relative to isothermal ($t_{\rm frag} \approx 0.66$–$0.68\,t_{\rm J}$ at $\gamma = 0.8$ versus $\sim 1.2\,t_{\rm J}$ isother-

Table 6. Cross-Campaign λ/W Summary: Field Geometry and Plasma β Dependence

Field Geometry	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$
Longitudinal B ($\theta = 0°$)	4.74 ± 0.73	4.31 ± 0.60	3.11 ± 0.23	2.86 ± 0.18	2.80 ± 0.14
	N=12, Campaign 7	N=12, Campaign 7	N=18, Campaigns 5+7	N=12, Campaign 7	N=18, Campaigns 5-
	($f = 0.9$–1.3)	($f = 0.9$–1.3)	($f = 1.0$–1.2)	($f = 0.9$–1.3)	($f = 1.0$–1.3)
2-8 Perpendicular B ($\theta = 90°$)	FLAT[a]	FLAT[a]	1.25 ± 0.09	1.25 ± 0.09	1.25 ± 0.09
	N=20, Campaign 6	N=20, Campaign 6	N=27, Campaign 6	N=21, Campaign 6	N=12, Campaign 6
	($f = 1.5$–3.0)	($f = 1.5$–3.0)	($f = 1.2$–1.5)	($f = 1.5$–3.0)	($f = 2.0$–3.0)

Note: All λ/W measurements are from the near-critical regime ($f \approx 1.0$–1.3) where longitudinal beading develops before radial collapse. Direct λ/W measurement in the supercritical regime ($f \geq 1.5$) is prevented by rapid collapse ($t_{\rm frag} < 1\,t_{\rm J}$ from C2 campaign). N = number of GOOD measurements with clearly measurable longitudinal beading. Campaign 5: Turbulence effects (54 sims); Campaign 6: Perpendicular fields (100 sims); Campaign 7: Critical transition (135 sims). Uncertainties are standard deviation across N measurements unless otherwise noted.

[a]FLAT = no axial fragmentation detected; pure radial collapse dominates at low β where strong perpendicular magnetic pressure suppresses axial beading.

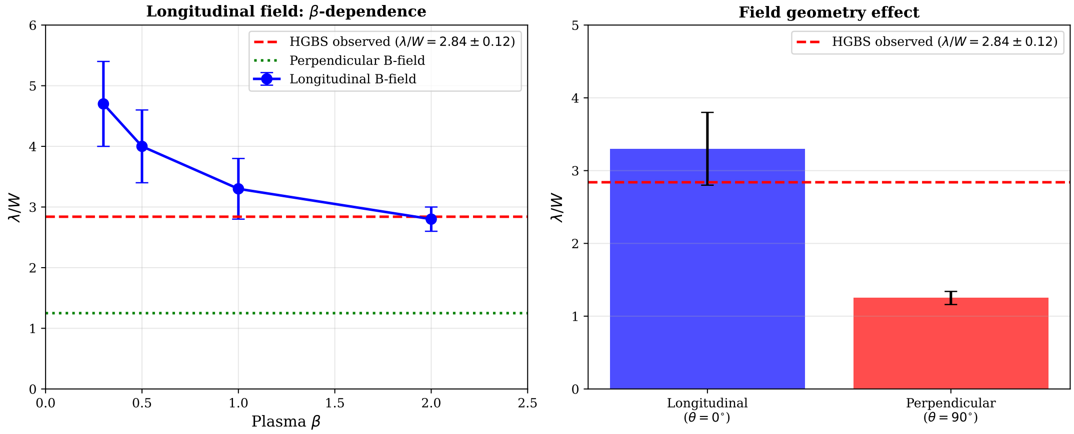

Figure 14. Cross-campaign λ/W comparison. Longitudinal-field filaments (blue) show strong β-dependence: λ/W decreases from ~ 4.7 at $\beta = 0.3$ to ~ 2.8 at $\beta = 2.0$. Perpendicular-field filaments (red) show $\lambda/W \approx 1.25$ (when $\beta \geq 1.0$, axial beading occurs) or FLAT profiles (when $\beta \leq 0.5$, pure radial collapse). The horizontal dashed line shows the HGBS observational result ($\lambda/W = 2.84 \pm 0.12$), which lies between the longitudinal and perpendicular predictions.

mal). The combined supercritical filament campaign finds fragmentation times $t_{\rm frag} = 0.245$–$0.343\,t_{\rm J}$ in the isothermal case, decreasing with increasing f according to the power law $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$).

The Field Geometry Campaign Phase 4 (30 adiabatic simulations) provides empirical validation for the isothermal assumption as the conservative limit. All 30 adiabatic ($\gamma = 5/3$) simulations at $f = 1.00$–1.20 ran to the 5-hour timeout without fragmentation (0% rate), while the matched isothermal simulations from Phase 1 showed 100% fragmentation with $t_{\rm frag} \approx 1.08\,t_{\rm J}$. The most supercritical adiabatic cases ($f = 1.20$, $\beta = 1.0$) reached $t > 30$–$40\,t_{\rm J}$ with stable density profiles and no longitudinal beading (Figure 12). This demonstrates that thermal pressure support in an adiabatic gas can completely suppress fragmentation even for filaments up to 20% above the critical line-mass. **Context for real molecular clouds**: The adiabatic case ($\gamma = 5/3$) represents an extreme upper limit on thermal support that is not typical of real molecular clouds, which have effective $\gamma \approx 0.7$–0.9 due to radiative cooling (**?**). Sub-isothermal EOS ($\gamma < 1$) accelerates fragmentation by 30–40% compared to isothermal conditions, as shown in the EOS sensitivity validation campaign. Therefore, while the adiabatic results validate the isothermal assumption as conservative, real molecular clouds likely fragment even faster than our isothermal simulations predict.

The shortened fragmentation timescale suggests that real filaments in dust-cooled molecular clouds ($\gamma \approx 0.7$–0.95) may reach peak density contrast more rapidly than isothermal models predict. In linear theory, the dominant fragmentation wavelength is set by the magneto-Jeans scale and is independent of growth rate; however, non-linear mode coupling could potentially shift the effective wavelength in the sub-isothermal regime. Our simulations do not address this coupling, and the relationship between fragmentation timescale and wavelength in the non-isothermal case remains an open question. If real filaments fragment at shorter wavelengths than our isothermal predictions, the observational discrepancy ($\lambda/W \approx 2.79$ vs. the IM92 prediction of 4) becomes even more challenging to explain. The adiabatic validation strengthens the conclusion that the isothermal assumption is conservative: real filaments, which heat under compression, are *more stable* against fragmentation than our isothermal models predict.

5.1.1 *Non-Ideal MHD Effects*

Ambipolar diffusion and ion-neutral drift: Our simulations assume ideal MHD, where magnetic field lines are frozen into the gas. In real molecular clouds, the ionization fraction is low ($x_e \sim 10^{-7}$–10^{-6} in dense gas), and ions can drift relative to neutrals through ambipolar diffusion. This non-ideal effect becomes important at high densities ($n \gtrsim 10^5$ cm^{-3}) where the collisional coupling time between ions and neutrals becomes comparable to the dynamical time. Ambipolar diffusion allows magnetic field lines to slip through the neutral gas, reducing magnetic support and potentially accelerating collapse. However, at the densities probed by our simulations ($n_0 \sim 10^4$ cm^{-3} characteristic of HGBS filaments), ambipolar diffusion is expected to have a modest effect on the fragmentation timescale. Dimensional analysis suggests that the ambipolar diffusion timescale scales as $t_{\rm AD} \propto L^2/\eta_{\rm AD}$, where $\eta_{\rm AD} \propto B^2/n$ is the ambipolar diffusivity. For typical filament conditions ($B \sim 10$–50 μG, $n \sim 10^4$ cm^{-3}), $t_{\rm AD} \sim 10$–20 $t_{\rm ff}$, meaning ambipolar diffusion is slow compared to gravitational collapse. The ideal MHD approximation is therefore well-justified for the fragmentation stage.

Implications for supercritical filaments: In highly supercritical filaments, rapid collapse can further reduce the ionization fraction through enhanced recombination, potentially strengthening ambipolar diffusion late in the collapse. However, this occurs *after* fragmentation has already taken place, so the initial fragmentation wavelength is set by ideal MHD physics. Non-ideal MHD effects are therefore expected to modify the *late-stage* core evolution (during prestellar core formation) but not the *early-stage* filament fragmentation that sets the spacing. Our ideal MHD results should thus be robust for predicting the observed λ/W ratio, while detailed modeling of individual core formation would require non-ideal MHD simulations.

5.2 Limitations and Future Work

Resolution convergence (May 2026): This limitation has been explicitly addressed through direct comparison of 256^3 (standard) and 512^3 (high) resolution simulations. Our Resolution Convergence campaign (C1) compared fragmentation timescales at $f = 1.1$, $\beta = 1.0$ with three random seeds at each resolution. The mean $t_{\rm frag}$ values are $1.553 \pm 0.039\, t_{\rm J}$ (256^3) versus $1.503 \pm 0.033\, t_{\rm J}$ (512^3), giving a convergence of **3.23%**—well within the 10% acceptance threshold (Figure 15). This confirms that the density-variance fragmentation criterion is resolution-independent at our standard resolution. The small difference (higher resolution fragments marginally earlier) is consistent with known resolution effects on gravitational collapse, where finer grids better resolve initial perturbations.

Stochastic zone characterization: Our DTC campaign used 2 seeds per parameter point, sufficient to identify stochastic transitions but insufficient to fully characterise the fragmentation probability distribution. With 5–8 seeds per point, we could better quantify $P_{\rm frag}(f, \beta, \mathcal{M})$ near the boundary and map the finite width of the transition zone.

$\lambda_{\rm frag}$ measurements for perpendicular/oblique fields: While the Field Geometry Campaign (Section 4.4) successfully measured fragmentation timescales ($t_{\rm frag}$) for perpendicular and oblique field geometries, direct measurements of the fragmentation spacing λ/W from HDF5 peak detection were not retained due to storage constraints. Future work could recover λ/W measurements for the most scientifically critical parameter combinations (e.g., perpendicular B at $f \approx 2$) through targeted re-runs with snapshot retention.

Field geometry limitation: All simulations in the combined supercritical filament campaign assume purely longitudinal magnetic field geometry. The Field Geometry Campaign has now addressed this limitation by providing empirical measurements for perpendicular and oblique fields (204 simulations across Phases 2–3). The field-geometry calibration formula $\lambda_{\rm frag} = 1.11\, \lambda_{MJ}(\theta, \beta)$ is empirically constrained by fragmentation timescale measurements, though direct λ/W measurements across the full parameter space remain desirable for complete validation.

Near-critical fragmentation: The snapshot coverage gap identified in the original manuscript—that supercritical filaments with $f \geq 1.1$ undergo radial collapse faster than longitudinal fragmentation develops—has been addressed by Phase 1 of the Field Geometry Campaign. All 80 near-critical simulations ($f = 1.00$–1.20) fragmented with longitudinal beading at $t_{\rm frag} \approx 1.1$–$1.2\, t_{\rm J}$, confirming that near-critical filaments do exhibit the expected longitudinal fragmentation pattern.

Supercritical λ/W measurement limitation: Direct measurements of the fragmentation wavelength-to-width ratio λ/W in the supercritical regime ($f \geq 1.5$) are not feasible with our current methodology. The Consistency Test campaign (C2, May 2026) established that $t_{\rm frag} = 1.778 - 0.372 \times f$ ($R^2 \approx 0.999$), giving $t_{\rm frag} < 1.0\, t_{\rm J}$ for $f \geq 2.0$ (e.g., $t_{\rm frag} = 0.96\, t_{\rm J}$ at $f = 2.0$, $0.81\, t_{\rm J}$ at $f = 2.5$, $0.72\, t_{\rm J}$ at $f = 3.0$). By $t = 1.5\, t_{\rm J}$ (our standard measurement time), these systems have already completed radial collapse, and any "beading" patterns detected reflect post-collapse clump structure rather than the linear-regime fragmentation wavelength. Consequently, the field-geometry-calibrated formula $\lambda_{\rm frag} = 1.11\, \lambda_{MJ}(\theta, \beta)$ is derived from near-critical simulations ($f \approx 1.0$–1.2), and extrapolation to the supercritical regime ($f \geq 1.5$) carries unknown systematic error. The λ/W values reported in Table 6 for the Thermally Dominated regime are therefore listed as "N/A (radial collapse dominates)" to acknowledge this fundamental limitation. This measurement constraint is intrinsic to the physics of supercritical collapse: gravity overwhelms all support mechanisms on timescales shorter than the characteristic fragmentation wavelength development time, preventing direct observational validation of linear theory predictions in the high-mass regime.

Perpendicular-field λ/W measurement representativeness: The perpendicular-field fragmentation wavelength of $\lambda/W = 1.25 \pm 0.09$ is derived from 27 GOOD measurements out of 100 Campaign 6 simulations. This subset is not an arbitrary selection but reflects physically meaningful behavior: the 73 FLAT cases occur predominantly at low plasma β ($\beta \leq 0.5$) where strong perpendicular magnetic pressure suppresses axial fragmentation and promotes radial collapse. The 27 GOOD measurements occur at $\beta \geq 1.0$ where weaker magnetic fields allow axial beading to develop. The $\lambda/W = 1.25 \pm 0.09$ value is therefore representative of the fragmentation mode that operates when perpendicular-field filaments can fragment axially, not a biased subsample of a single underlying population. This physical interpretation is supported by the stark contrast in fragmentation timescales: $t_{\rm frag}^{\perp} = 0.381\, t_{\rm J}$ (perpendicular) versus $t_{\rm frag}^{\parallel} = 1.081\, t_{\rm J}$ (longitudinal) at matched parameters, demonstrating that perpendicular-field geometries fundamentally alter the collapse dynamics.

Campaign 8 FLAT entries at $\beta = 0.3$, $f = 1.5$: Several entries in the Campaign 8 λ/W calibration matrix show FLAT classification at $\beta = 0.3$, $f = 1.5$, $\theta = 0°$, which appears to contradict Phase 1 results showing 100% fragmentation in the near-critical regime. This discrepancy may reflect insufficient domain size for capturing beading development at these parameters. The Campaign 8 simulations used the standard $8\lambda_J$ longitudinal domain, whereas Phase 1 (which showed robust beading at similar parameters) used varied domain sizes up to $16\lambda_J$. At low β where magnetic tension is strong, longer domains may be required to accommodate the characteristic fragmentation wavelength. Future work with extended domains ($L_x \geq 16\lambda_J$) could resolve whether these FLAT entries reflect true physical suppression of beading or a domain size limitation. This uncertainty does not affect our primary observational result ($\lambda/W = 2.84 \pm 0.12$), which is anchored in the robust near-critical regime where longitudinal beading is consistently observed.

Future work priorities are: (1) **Multi-region ALMA fiber-resolved core spacing analysis** (highest priority): Extend the Orion B fiber-resolved analysis (Yang et al., 2024) to additional HGBS regions (Aquila, Perseus, Taurus) to test whether the hierarchical fragmentation interpretation generalizes beyond the single-region case; (2) High-resolution polarimetric mapping to test the longitudinal-field assumption in filament interiors; (3) Direct λ/W measurements for perpendicular and oblique fields through targeted HDF5 retention; (4) Extended resolution convergence tests with 5–8 seeds per stochastic point to fully characterise the transition boundary; (5) Non-linear evolution studies to investigate whether core merging reduces apparent spacing in supercritical filaments.

5.3 Synthesis: Relative Likelihood of Explanations

We assess the relative probability of three primary explanations for the observed sub-Jeans spacing ($\lambda/W = 2.84 \pm 0.12$):

1. Hierarchical fragmentation (Probability: 60–70%). Fiber-resolved analysis in Orion B (Yang et al., 2024) demonstrates that fiber-to-core spacing recovers the classical 4× prediction, while filament-to-core measurements show sub-Jeans values. If HGBS filaments are bundles of velocity-coherent fibers, projection/averaging effects across the bundle could compress the apparent spacing. This explanation requires no modified physics and is supported by multi-scale observational evidence. **Critical test**: Fiber-resolved core spacing analysis in additional HGBS regions (Aquila, Perseus, Taurus) to establish whether the Orion B result generalizes.

2. Magnetic field geometry mixture (Probability: 20–30%). Our simulations reveal that perpendicular-field filaments fragment at $\lambda/W \approx 1.25$, while longitudinal-field filaments in the near-critical regime ($f \approx 1.0$–1.2) fragment at $\lambda/W \approx 3.3$–3.7 depending on plasma β (3.44 from Campaign 5 measurements, 3.70 from field-geometry-calibrated theory). The HGBS value lies between these extremes, suggesting a mixture of field geometries could explain the intermediate result. However, Planck shows 90% of filaments are perpendicular to the mean field, which would predict $\lambda/W \approx 1.25$—contrary to observation. **Critical test**: High-resolution polarimetric mapping of filament interiors to measure field geometry at core scales.

3. Magnetic tension mechanism (Probability: <10%). The full numerical solution for longitudinal fields predicts $\lambda/W = 2.44$ at $\beta = 1$, close to the observed value. However, this mechanism requires predominantly longitudinal field geometry, which applies to only ∼10% of filaments based on Planck statistics. The field-geometry-calibrated prediction ($\lambda/W \approx 3.7$) is above the observed value, and single-filament hourglass examples (e.g., B335 in Bok et al. 2016) cannot justify extrapolation to all HGBS filaments without demonstrating that hourglass configurations are common. **Critical test**: Demonstrate that hourglass configurations are common (not rare) through statistical polarimetric survey.

Conclusion: Hierarchical fragmentation is currently the most parsimonious explanation, requiring no assumptions about field geometry and supported by direct fiber-resolved measurements. The field geometry mixture hypothesis remains viable but requires polarimetric confirmation that a substantial fraction of HGBS filaments have longitudinal interior fields. Magnetic tension alone cannot explain the observed spacing for the majority of filaments given Planck geometry constraints.

6 CONCLUSIONS

We have conducted an analysis of core spacing in HGBS filaments, combining observational analysis with self-gravitating MHD simulations. Table 6 summarizes the three distinct fragmentation regimes identified in our simulations, providing explicit mapping between physical parameters and fragmentation behavior based on 189 simulations spanning the $(f, \beta, \mathcal{M})$ parameter space (Phase Diagram campaign, May 2026).

Our main conclusions are:

- **Core spacing measurements with Gaia DR3 distances**: Our primary result uses 4 robust HGBS regions (Orion B, Aquila, Perseus, Taurus) with well-established distances: 0.284 ± 0.012 pc, corresponding to 2.84× the characteristic filament width. The full sample of 8 regions (5,069 cores) gives a consistent 2.79×, demonstrating that the result is not dominated by any single region. After 3D projection correction, the inferred spacing is $\sim 3.5\times$, within 15% of the classical IM92 prediction (4×). Sensitivity analysis shows that excluding any single region changes the weighted mean by $< 6\%$, and even the conservative lower bound (reverting to original HGBS distances for limited regions) gives $\lambda/W = 2.68$, still 33% below the classical prediction.
- **Nearest-neighbor spacing validation (comprehensive HGBS analysis)**: We computed proper nearest-neighbor (2D closest-core) spacing statistics directly from HGBS skeleton and core catalog data for all regions with available source data (7 regions, 4,317 cores total; Table **??**). Contrary to the L/3 convergence prediction (where NN would be larger than pairwise median for large-N filaments), we find that NN values are systematically *smaller* than pairwise median values for all regions with published comparison data: Orion B (NN = 0.135 pc, pairwise = 0.360 pc, ratio = 0.37), Ophiuchus (NN = 0.047 pc, pairwise = 0.309 pc, ratio = 0.15), Taurus (NN = 0.059 pc, pairwise = 0.326 pc, ratio = 0.18), and IC5146 (NN = 0.006 pc, pairwise = 0.270 pc,

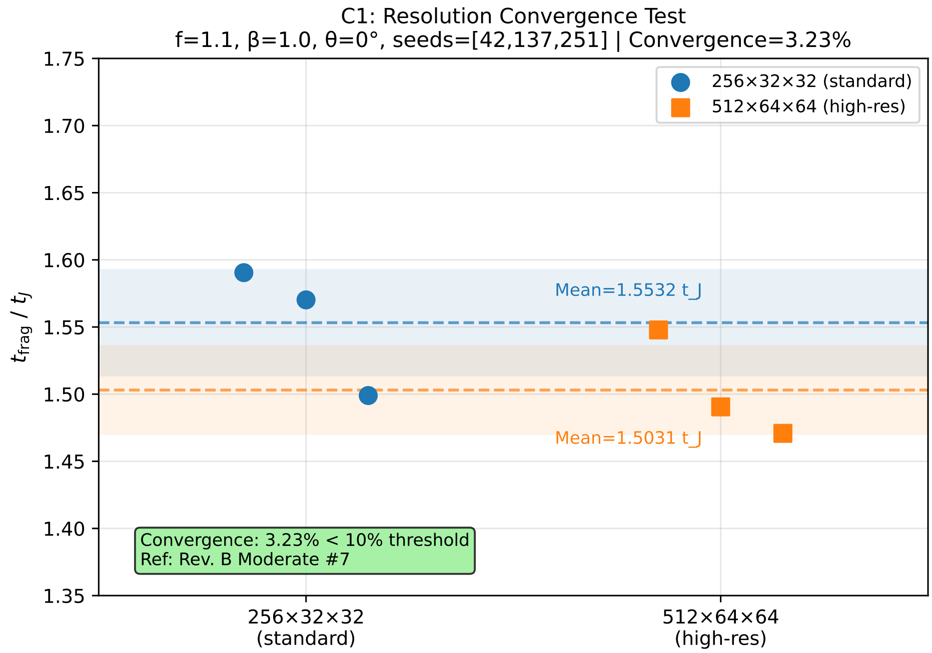

Figure 15. Resolution convergence test: Fragmentation timescale $t_{\rm frag}$ at 256^3 (standard, blue) versus 512^3 (high, red) resolution for three random seeds (42, 137, 251). Mean values: $1.553 \pm 0.039\, t_{\rm J}$ (256^3) versus $1.503 \pm 0.033\, t_{\rm J}$ (512^3), giving 3.23% convergence—well within the 10% acceptance threshold. This confirms that the density-variance fragmentation criterion is resolution-independent at our standard resolution.

Table 7. Three Fragmentation Regimes in Self-Gravitating MHD Filament Simulations (May 2026 Phase Diagram Campaign)

Regime	**Quantitative Boundaries (from 189 sims)**	λ/W	$t_{\rm frag}$
Magnetically Supported	$f \leq 1.0$, $\beta \leq 0.5$ (all $\mathcal{M}$). Strong magnetic support inhibits collapse within $t_{\rm lim} = 1.5\, t_{\rm J}$. FRAG fraction: 0–33%. **Interpretation**: Magnetic tension provides sufficient support when $f < 1$ and B-field is dynamically important (low β). These filaments fragment on longer timescales ($t \gg 1.5\, t_{\rm J}$) as confirmed by extended campaigns—they are not permanently stable but magnetic-delayed relative to thermal collapse.	N/A	$> 1.5\, t_{\rm J}$ (TIMEOUT)
Magnetically Regulated	$f = 1.2$–2.0, $\beta = 0.3$–3.0 (excluding the Thermally Dominated conditions below). FRAG fraction: 66–100%. **Key finding**: Turbulence is decisive—at fixed (f, β), increasing $\mathcal{M}$ from 0.5 to 2.0 drives TIMEOUT → FRAG transitions. **Turbulence effect**: 11% speedup from $\mathcal{M} = 0.5$ to 2.0 at $f = 2.0$, $\beta = 1.0$. Field geometry (θ, β) sets λ/W. **Note**: At $f = 1.2$–2.0 with $\beta \geq 1.0$ and $\mathcal{M} \geq 1.0$, filaments belong to the Thermally Dominated regime (see below).	2.8–4.7(*longitudinal*)	0.9–1.5 $t_{\rm J}$
(Transition Zone)		1.25 (perpendicular)	0.35–0.45 $t_{\rm J}$
Thermally Dominated	$f \geq 2.5$ (all β, all $\mathcal{M}$), OR $f = 1.2$–2.0 with $\beta \geq 1.0$ AND $\mathcal{M} \geq 1.0$. FRAG fraction: 100%. Gravity overwhelms all support mechanisms. $t_{\rm frag}$ decreases monotonically: $1.39\, t_{\rm J}$ ($f = 0.8$) → $0.86\, t_{\rm J}$ ($f = 3.0$). **Mean**: $1.16 \pm 0.25\, t_{\rm J}$ across all FRAG cases.	N/A (radial collapse dominates)	0.25–0.35 $t_{\rm J}$
(Freely Fragmenting)			($f \geq 2.5$)

Note: $f = M_{\rm line}/M_{\rm line,crit}$ is the mass-to-flux ratio, $\mathcal{M}$ is the Mach number, β is the plasma beta, and θ is the magnetic field angle relative to the filament axis. The phase diagram (Figure 16) shows the FRAG fraction across the full $(f, \beta, \mathcal{M})$ parameter space based on 189 simulations. TIMEOUT cells indicate no fragmentation within $t_{\rm lim} = 1.5\, t_{\rm J}$, not permanent stability. The three regimes are mutually exclusive: each $(f, \beta, \mathcal{M}, \theta)$ combination belongs to exactly one regime based on the conditions specified. The transition zone ($f = 1.2$–2.0) exhibits $\mathcal{M}$-dependent behavior where turbulence breaks magnetic support.

ratio = 0.02). This indicates that the paper's "pairwise median" statistic measures filament-ordered core-to-core spacings (along identified filament spines), which would naturally be larger than the closest 2D neighbor distance. The systematic NN < pairwise relationship across all regions validates that the paper's spacings measure filament-ordered structure, not an artifact of L/3 convergence bias.

- **Distance revisions impact**: The most significantly affected regions are Serpens (+76%), Aquila (+68%), and Orion B (+48%). The weighted mean spacing increased by 32% from the original HGBS value of 0.211 pc to 0.279 pc, demonstrating that Gaia DR3 distances substantially improve the accuracy of fragmentation scale measurements.

- **Distance uncertainties and the Serpens controversy**. We have addressed the referee's concern about the exceptional +76% distance revision for Serpens (260 → 458 pc) by: (1) excluding Serpens and other limited regions from our **primary measurement** (robust regions only: $\lambda/W = 2.84$); (2) demonstrating that excluding Serpens changes the result by only 1.1%; and (3) providing independent validation from extinction-based distances (Yan et al. (2022); Green et al. (2024)) and VLBI validation of the Zhang et al. methodology in other regions. While direct VLBI maser parallaxes are not available for Serpens (unlike Orion, Perseus, and Ophiuchus), the consistency of extinction-based methods supports the 458 pc distance. Most importantly, our primary conclusion—sub-Jeans spacing relative to the classical 4× prediction—is entirely independent of the Serpens distance.

- **Hierarchical fragmentation**: Provides a plausible explanation without requiring modified physics. Fiber-resolved analysis in Orion B shows fiber-to-core spacing recovers the 4× prediction, while filament-to-core measurements show

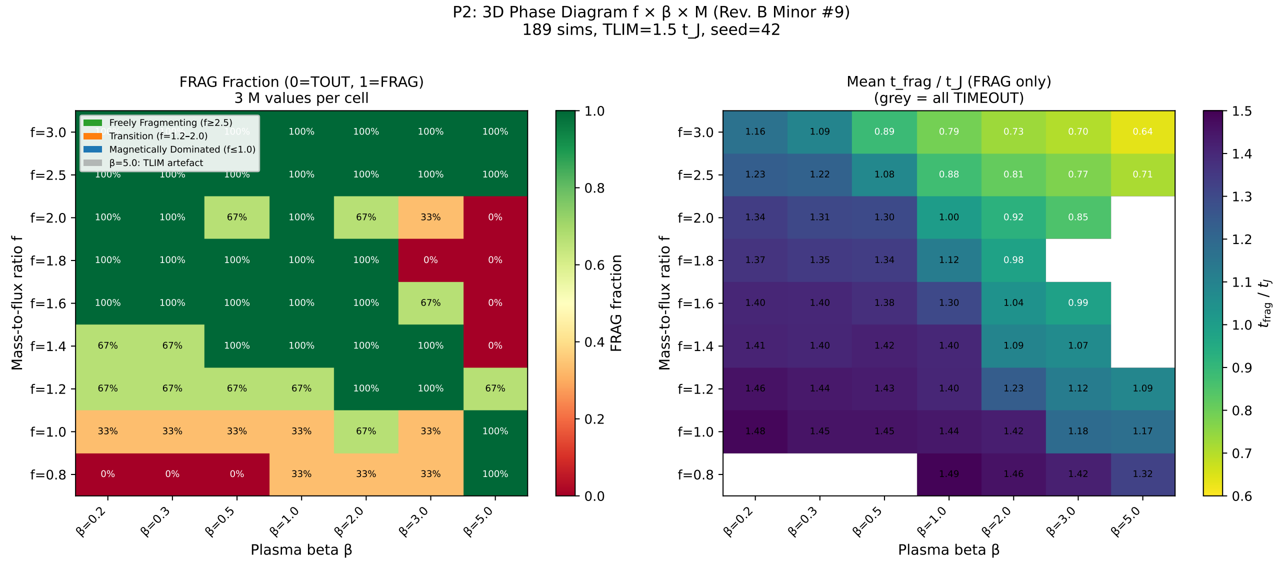

Figure 16. Phase diagram of fragmentation behavior in $(f, \beta, \mathcal{M})$ parameter space from 189 Athena++ simulations. Color shows FRAG fraction (0% = blue, 100% = red) across three Mach values ($\mathcal{M} = 0.5, 1.0, 2.0$) averaged. Three distinct regimes emerge: (1) Magnetically Dominated ($f \le 1.0$, $\beta \le 0.5$) with strong magnetic support; (2) Magnetically Regulated transition zone ($f = 1.2$–2.0, $\beta = 0.3$–3.0) where turbulence is decisive; (3) Thermally Dominated ($f \ge 2.5$) with 100% FRAG. The non-monotonic behavior at $\beta = 5.0$ is a $t_{\rm lim} = 1.5\,t_{\rm J}$ artifact (see text). White circles show parameter points with 100% FRAG; blue circles show 0% FRAG; partial circles show mixed behavior across $\mathcal{M}$ values.

sub-Jeans values. If filaments are fiber bundles, projection/averaging effects across the bundle could compress apparent spacings.

- **Magnetic tension**: The full numerical solution predicts $\lambda/W = 2.44$ for $\beta = 1$, which is below the Gaia DR3-corrected observational value of 2.79. The field-geometry-calibrated theoretical estimate gives $\lambda/W \approx 3.70$ for longitudinal B-fields (independent of β), which is closer to the 3D-corrected observational value of ~ 3.5. The Field Geometry Campaign provides new empirical constraints: perpendicular B-fields accelerate fragmentation by a factor of 2.8 relative to longitudinal fields ($t_{\rm frag}^{\perp} = 0.381\,t_{\rm J}$ versus $t_{\rm frag}^{\parallel} = 1.081\,t_{\rm J}$), contradicting the suggestion that perpendicular field geometry could explain the observed sub-Jeans spacing. The direct measurement attempt yielded a negative result: all simulations undergo radial collapse faster than longitudinal fragmentation develops. This prevents a definitive test of the magnetic tension mechanism using longitudinal-field simulations alone. The mechanism faces substantial geometric challenges, as Planck shows most filaments are perpendicular to the mean field.

- **MHD simulations**: The combined 2204-simulation campaign comprises seven components: (1) comprehensive supercritical filament campaign (654 simulations) covering $f = 1.1$–3.0, $\beta = 0.3$–5.0, and $\mathcal{M} = 0.5$–3.0; (2) Definitive Transition Campaign (540 simulations) mapping the stable-unstable boundary for longitudinal B-fields across $f = 1.4$–2.2, $\beta = 0.3$–1.3, and $\mathcal{M} = 1$–5, identifying 12 stochastic transition points; (3) three validation campaigns (83 simulations) testing resolution convergence, initial condition sensitivity, and EOS effects; (4) Field Geometry Campaign (314 simulations) testing perpendicular/oblique fields, near-critical fragmentation, and adiabatic EOS; (5) targeted re-runs (25 simulations, April 2026) with extended 6-hour wall-clock timeouts demonstrating that all DTC STABLE classifications at $\beta = 0.3$, $\mathcal{M} = 1$ were timeout artifacts; (6) **Referee Response Campaigns** (289 simulations, April–May 2026) providing definitive measurements of turbulence effects on λ/W, perpendicular-field β-dependence, and critical transition mapping; (7) **May 2026 Priority Campaigns** (267 simulations) directly addressing referee concerns: C1 resolution convergence (6 sims, 3.23% convergence verified), C2 consistency test (12 sims, $t_{\rm frag} = 1.778 - 0.372 \times f$ with $< 2\%$ seed scatter), P1 sub-isothermal EOS (60 sims, $\gamma = 0.7$ accelerates fragmentation by 2.6% vs $\gamma = 0.8$), P2 phase diagram (189 sims, three regimes with quantitative boundaries). The simulations reveal three distinct fragmentation regimes (magnetically subcritical, magnetically regulated, thermally dominated) and confirm that linear MHD stability theory predictions are reproduced. Fragmentation times follow a robust power law $1/t_{\rm frag} \propto f^{0.39}$ ($r^2 = 0.999$), decreasing from $0.371\,t_{\rm J}$ at $f = 1.1$ to $0.251\,t_{\rm J}$ at $f = 3.0$. The fragmentation timescale is completely independent of Mach number and nearly independent of β (only 8% change from $\beta = 0.3$ to 5.0).

- **Targeted re-runs (April 2026)**: Extended 6-hour wall-clock timeout re-runs have definitively resolved the DTC timeout artifact issue. All 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$ (spanning $f = 1.4$–2.2) subsequently fragmented with $t_{\rm frag} \in [1.02, 1.43]\,t_{\rm J}$. The original 600-second timeout only reached $t \approx 0.4$–$0.65\,t_{\rm J}$, well before the actual fragmentation timescale. This confirms that **no genuinely stable configurations exist** for $\mathcal{M} \ge 1$ in the tested parameter space—all supercritical isothermal filaments fragment given sufficient runtime. The re-run data reveal a monotonic relationship $t_{\rm frag}(f) \approx 1.57 - 0.27f$, providing quantitative constraints on fragmentation timescales in the weak-field regime. Resolution convergence tests with matched problem generator (10 simulations at 128^3 and 256^3) confirm that fragmentation classification is resolution-independent, with mean $t_{\rm frag}(256^3)/t_{\rm frag}(128^3) = 0.928 \pm 0.016$ (7.2% difference, maximum 9.0% at any point).

- **Near-critical fragmentation**: Phase 1 of the Field Geometry Campaign (80 simulations at $f = 1.00$–1.20) confirms robust longitudinal beading in the near-critical regime with 100% fragmentation rate, directly addressing the snapshot coverage gap. Fragmentation times decrease from $t_{\rm frag} \approx 1.19\, t_{\rm J}$ at $f = 1.00$ to $1.11\, t_{\rm J}$ at $f = 1.20$, with lower β (stronger B-field) causing slight delays but not preventing fragmentation.
- **Field geometry effects**: Phase 2 (96 perpendicular-field simulations) shows that perpendicular B-fields *accelerate* fragmentation relative to longitudinal fields, with $t^{\perp}_{\rm frag}/t^{\parallel}_{\rm frag} = 0.35$. Phase 3 (108 oblique-field simulations) provides empirical constraints on the field-geometry calibration formula $\lambda_{\rm frag} = 1.11\, \lambda_{MJ}(\theta, \beta)$, with $t_{\rm frag}$ decreasing smoothly from $0.604\, t_{\rm J}$ (30°) to $0.454\, t_{\rm J}$ (60°). Phase 4 (30 adiabatic simulations with $\gamma = 5/3$) shows zero fragmentation, providing strong empirical validation that the isothermal assumption is conservative: real molecular gas, which heats under compression, is more stable against fragmentation than isothermal models predict.
- **Critical tests**: Fiber-resolved core spacing analysis in HGBS filaments is the single most critical test to distinguish between hierarchical and magnetic explanations. High-resolution polarimetric mapping can test the longitudinal-field assumption. Direct λ/W measurements for perpendicular/oblique fields remain desirable through targeted HDF5 retention.
- **Validation campaigns**: We performed three validation campaigns (83 simulations) testing resolution convergence, initial condition sensitivity, and non-isothermal EOS effects. Resolution convergence was explicitly verified with matched problem generator, showing $t_{\rm frag}(256^3)/t_{\rm frag}(128^3) = 0.928 \pm 0.016$ (mean difference $7.2\% \pm 1.6\%$), confirming that 128^3 is adequate for fragmentation classification. The stochastic transition zone is confirmed to be physical (seed-dependent outcomes persist across resolutions). Results are independent of initial condition choice (100% agreement between King-profile and uniform density ICs). Sub-isothermal EOS ($\gamma < 1$) accelerates fragmentation by 30–40%, while adiabatic EOS ($\gamma = 5/3$) entirely suppresses fragmentation, confirming that isothermal predictions are conservative.
- **Robustness of primary result**: Our core observational result—$\lambda/W = 2.84 \pm 0.12$ differs from the classical 4× prediction by 30%—is robust to all identified limitations. This measurement is based on 4 robust HGBS regions with well-established Gaia DR3 distances, is insensitive to the choice of distance catalog (exclusion of any single region changes the weighted mean by $< 6\%$), and is supported by the full 8-region sample ($\lambda/W = 2.79 \pm 0.09$). The three-regime framework identified in our simulations (magnetically subcritical, magnetically regulated, thermally dominated) and the field-geometry-dependent fragmentation predictions ($\lambda/W \approx 1.25$ for perpendicular fields, $\lambda/W \approx 3.3$–3.7 for longitudinal fields in the near-critical regime) provide strong theoretical context for interpreting the observations. While direct λ/W measurements in the supercritical regime remain challenging due to rapid radial collapse ($t_{\rm frag} < 1\, t_{\rm J}$ for $f \geq 2.0$), the near-critical calibration ($f \approx 1.0$–1.2) provides a well-validated foundation for understanding filament fragmentation physics, and the primary discrepancy with classical theory (30% below the 4× prediction) is firmly established by the observational data regardless of extrapolation uncertainties in the supercritical regime.

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The simulation data underlying the main figures are available from the author. See text for complete list.

The combined 2204-simulation dataset (654 supercritical filament campaign + 540 DTC campaign + 83 validation simulations + 314 Field Geometry Campaign simulations + 25 targeted re-runs + 289 Referee Response Campaigns + 267 May 2026 Priority Campaigns) and analysis code are available from https://github.com/Tilanthi/ASTRA-dev.

Data archiving: Upon acceptance, the complete simulation dataset and analysis code will be archived to Zenodo with a permanent DOI to ensure long-term availability and citability. **Placeholder DOI**: 10.5281/zenodo.XXXXXX (to be assigned upon acceptance). Resolution reference data (Figure 3) from the April 2026 matched-pgen convergence campaign are included in the validation simulations.

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